

UNIVERSIDADE FEDERAL DE SANTA CATARINA
CAMPUS REITOR JOÃO DAVID FERREIRA LIMA
GRADUATE PROGRAM IN COMPUTER SCIENCE

Gabriel de Oliveira Aguiar

Crash-Aware Traffic Prioritization in Vehicular Wi-Fi Networks

Florianópolis
2026

Gabriel de Oliveira Aguiar

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Master's Dissertation submitted to the Graduate Program in Computer Science of the Universidade Federal de Santa Catarina as a requirement for obtaining the degree of Master of Science in Computer Science.
Supervisor: Prof. Giovani Gracioli, Ph.D.

Florianópolis
2026

Ficha de identificação da obra elaborada pelo autor,
através do Programa de Geração Automática da Biblioteca Universitária da UFSC.

Aguiar, Gabriel de Oliveira
Projeto e Desenvolvimento de um Simulador de Irradiância Solar para CubeSats : Uma Alternativa de Baixo Custo / Gabriel de Oliveira Aguiar ; orientador, Eduardo Augusto Bezerra, 2023.
59 p.

Trabalho de Conclusão de Curso (graduação) - Universidade Federal de Santa Catarina, Centro Tecnológico, Graduação em Engenharia Eletrônica, Florianópolis, 2023.

Inclui referências.

1. Engenharia Eletrônica. 2. CubeSats. 3. Satélites. I. Bezerra, Eduardo Augusto . II. Universidade Federal de Santa Catarina. Graduação em Engenharia Eletrônica. III. Título.

Gabriel de Oliveira Aguiar

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This Master's work was evaluated and approved by an examination committee composed of the following members:

Prof.(a) xxxx, Dr(a).
Institution xxxx

Prof.(a) xxxx, Dr(a).
Institution xxxx

Prof.(a) xxxx, Dr(a).
Institution xxxx

We certify that this is the **original and final version** of the thesis that was deemed suitable for obtaining the degree of Master of Science in Computer Science.

Graduate Program Coordination

Prof. Giovani Gracioli, Ph.D.
Supervisor

Florianópolis, 2026.

RESUMO

Alertas de acidente em comunicação veículo-a-tudo (*Vehicle-to-Everything*, V2X) são mensagens raras, porém de alto impacto, cuja utilidade depende de entrega oportuna sob contenção de canal. Em redes IEEE 802.11p descentralizadas, esses alertas compartilham o meio com o tráfego de rotina, de modo que picos de atraso e congelamento de *backoff* podem corroer sua prioridade sob carga. Esta dissertação avalia uma estratégia focada de priorização ciente de acidente na qual todos os veículos trocam tráfego *multicast* periódico de melhor esforço (*Best Effort*, BE), enquanto um veículo acidentado emite rajadas temporárias de alerta marcadas com *Differentiated Services Code Point* (DSCP) 46 e mapeadas para a categoria de voz (*Voice*, VO). Usando OMNeT++, Veins e INET, cinco políticas de controle de acesso ao meio (MAC) são comparadas em cenários de rodovia com 10 e 100 veículos: uma linha de base com *Distributed Coordination Function* (DCF), *Enhanced Distributed Channel Access* (EDCA) com mapeamento explícito de DSCP para voz, e três extensões adaptativas que temporariamente adiam ou descartam o tráfego BE concorrente enquanto os alertas estão ativos, todas sem exigir mudanças de protocolo ou infraestrutura. Sob um oferecimento Poisson uniforme de Best Effort (média de 125 ms), a idade de alerta no percentil 95 permanece em cerca de 2,2 ms nas cinco políticas MAC: o bloqueio adaptativo não reduz essa cauda, e o bloqueio sem descarte onera o Best Effort (`stable` eleva o P95 de BE da frota de 0,51 ms para 30 ms) sem comprar serviço de Voz. Sob contenção concentrada próxima ao acidente—poucos remetentes volumosos adjacentes—com rajadas de alerta densas, a preempção de emergência recupera o serviço de Voz: média 39% menor e percentil 95 21% menor de idade de alerta que a linha de base DCF, com 30% menos perdas de Voz por colisão de quadros que o EDCA puro. Os resultados caracterizam uma fronteira proteção-custo condicional ao regime de contenção: o bloqueio adaptativo só se justifica como escalada acionada por eventos sob contenção concentrada, e não como um modo de QoS permanentemente ativo.

Palavras-chave: V2X. Wi-Fi Veicular. IEEE 802.11p. EDCA. Qualidade de Serviço. Priorização Ciente de Acidente.

ABSTRACT

Vehicle-to-everything (V2X) crash alerts are rare but high-impact messages whose usefulness depends on timely delivery under channel contention. In decentralized IEEE 802.11p networks they share the medium with routine traffic, so delay spikes and back-off freezing can erode their priority under load. This dissertation evaluates a focused crash-aware prioritization strategy in which all vehicles exchange periodic Best Effort (BE) multicast traffic while one crash vehicle emits temporary alert bursts marked with Differentiated Services Code Point (DSCP) 46 and mapped to Voice (VO). Using OMNeT++, Veins, and INET, five Medium Access Control (MAC) policies are compared on highway scenarios with 10 and 100 vehicles: a Distributed Coordination Function (DCF) baseline, Enhanced Distributed Channel Access (EDCA) with explicit DSCP-to-Voice mapping, and three adaptive extensions that temporarily suppress or drop competing BE traffic while alerts are active, all requiring no protocol changes or infrastructure. With a uniform Poisson Best Effort offer (mean 125 ms), crash-alert 95th-percentile age stays about 2.2 ms under all five MAC policies: adaptive blocking does not improve that tail, and blocking without dropping taxes Best Effort (stable raises fleet BE P95 from 0.51 ms to 30 ms) without buying Voice service. Under concentrated near-crash contention—a few crash-adjacent bulk senders—with dense alert bursts, emergency preemption recovers Voice service: 39% lower mean and 21% lower 95th-percentile alert age than the DCF baseline, with 30% fewer Voice collision losses than plain EDCA. The results characterize a regime-conditional protection-versus-cost frontier: adaptive blocking is justified only as an event-triggered escalation under concentrated contention, not as an always-on QoS mode.

Keywords: V2X. Vehicular Wi-Fi. IEEE 802.11p. EDCA. Quality of Service. Crash-Aware Prioritization.

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LIST OF ABBREVIATIONS AND ACRONYMS

3GPP	3rd Generation Partnership Project (*Projeto de Parceria de 3ª Geração*)
AC	Access Category (*categoria de acesso*)
ACK	Acknowledgment (*confirmação de recebimento*)
AIFS	Arbitration Interframe Space (*espaço entre quadros de arbitragem*)
AIFSN	Arbitration Interframe Space Number (*número do espaço entre quadros de arbitragem*)
AP	Access Point (*ponto de acesso*)
BE	Best Effort (*melhor esforço*)
BEB	Binary Exponential Backoff (*retirada exponencial binária*)
BK	Background (*categoria background*)
BSM	Basic Safety Message (*mensagem básica de segurança*)
BSS	Basic Service Set (*conjunto básico de serviços*)
CAM	Cooperative Awareness Message (*mensagem de consciência cooperativa*)
CBR	Channel Busy Ratio (*razão de ocupação do canal*)
CCH	Control Channel (*canal de controle*)
CSMA/CA	Carrier Sense Multiple Access with Collision Avoidance (*acesso múltiplo por detecção de portadora com prevenção de colisões*)
CTS	Clear to Send (*autorização para enviar*)
CW	Contention Window (*janela de contenção*)
DCC	Decentralized Congestion Control (*controle de congestionamento descentralizado*)
DCF	Distributed Coordination Function (*função de coordenação distribuída*)
DENM	Decentralized Environmental Notification Message (*mensagem de notificação ambiental descentralizada*)
DiffServ	Differentiated Services (*serviços diferenciados*)
DIFS	DCF Interframe Space (*espaço entre quadros DCF*)
DSCP	Differentiated Services Code Point (*ponto de código de serviços diferenciados*)
DSRC	Dedicated Short-Range Communications (*comunicações dedicadas de curto alcance*)
EDCA	Enhanced Distributed Channel Access (*acesso ao canal distribuído aprimorado*)
EDCF	Enhanced Distributed Coordination Function (*função de coordenação distribuída aprimorada*)

EF	Expedited Forwarding (*encaminhamento acelerado*)
ETSI	European Telecommunications Standards Institute (*Instituto Europeu de Normas de Telecomunicações*)
FSM	Finite-State Machine (*máquina de estados finitos*)
GCR	Groupcast with Retries (*retransmissão de grupo confiável*)
HCCA	HCF Controlled Channel Access (*acesso ao canal controlado por HCF*)
HCF	Hybrid Coordination Function (*função de coordenação híbrida*)
IEEE	Institute of Electrical and Electronics Engineers (*instituto de engenheiros eletricistas e eletrônicos*)
INET	INET Framework (*framework de redes do OMNeT++*)
IP	Internet Protocol (*protocolo de internet*)
IPv4	Internet Protocol version 4 (*protocolo de internet versão 4*)
IPv6	Internet Protocol version 6 (*protocolo de internet versão 6*)
ITS	Intelligent Transportation Systems (*sistemas inteligentes de transporte*)
IVC	Inter-Vehicle Communication (*comunicação entre veículos*)
KPI	Key Performance Indicator (*indicador-chave de desempenho*)
MAC	Medium Access Control (*controle de acesso ao meio*)
MIMO	Multiple-Input Multiple-Output (*múltiplas entradas e múltiplas saídas*)
MSDU	MAC Service Data Unit (*unidade de dados de serviço MAC*)
NED	Network Description (*descrição de rede do OMNeT++*)
OCB	Outside the Context of a BSS (*fora do contexto de um BSS*)
OFDM	Orthogonal Frequency-Division Multiplexing (*multiplexação ortogonal por divisão de frequência*)
OFDMA	Orthogonal Frequency-Division Multiple Access (*acesso múltiplo por divisão de frequência ortogonal*)
OMNeT++	Objective Modular Network Testbed in C++ (*simulador de redes de eventos discretos*)
P95	95th percentile (*percentil 95*)
PHB	Per-Hop Behavior (*comportamento por salto*)
PHY	Physical Layer (*camada física*)
QoS	Quality of Service (*qualidade de serviço*)
RFC	Request for Comments (*pedido de comentários*)
RSU	Roadside Unit (*unidade de beira de estrada*)
RTS	Request to Send (*solicitação para enviar*)
SCH	Service Channel (*canal de serviço*)
SIFS	Short Interframe Space (*espaço entre quadros curto*)

SNIR	Signal-to-Noise-plus-Interference Ratio (*relação sinal-ruído-mais-interferência*)
SPS	Semi-Persistent Scheduling (*agendamento semi-persistente*)
SUMO	Simulation of Urban MObility (*simulação de mobilidade urbana*)
TAS	Time-Aware Shaper (*modelador ciente do tempo*)
TDMA	Time Division Multiple Access (*acesso múltiplo por divisão de tempo*)
TPC	Transmit Power Control (*controle de potência de transmissão*)
TraCI	Traffic Control Interface (*interface de controle de tráfego*)
TRC	Transmit Rate Control (*controle de taxa de transmissão*)
TSN	Time-Sensitive Networking (*redes sensíveis ao tempo*)
TXOP	Transmission Opportunity (*oportunidade de transmissão*)
UAV	Unmanned Aerial Vehicle (*veículo aéreo não tripulado*)
UDP	User Datagram Protocol (*protocolo de datagrama do usuário*)
UP	User Priority (*prioridade de usuário*)
URLLC	Ultra-Reliable Low-Latency Communication (*comunicação ultraconfiável de baixa latência*)
V2V	Vehicle-to-Vehicle (*comunicação veículo-a-veículo*)
V2X	Vehicle-to-Everything (*comunicação veículo-a-tudo*)
VANET	Vehicular Ad hoc Network (*rede ad hoc veicular*)
Veins	Vehicles in Network Simulation (*framework de simulação veicular*)
VI	Video (*vídeo*)
VO	Voice (*voz*)
WAVE	Wireless Access in Vehicular Environments (*acesso sem fio em ambientes veiculares*)

LIST OF SYMBOLS

W	Current contention-window size
CW_{min}	Minimum contention window
CW_{max}	Maximum contention window
p	Conditional collision probability
τ	Per-slot transmission probability
m	Maximum backoff stage
AIFS[i]	Arbitration interframe space of access category i
SIFS	Short interframe space duration
AIFSN[i]	AIFSN parameter of access category i
aSlotTime	Slot time

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1 INTRODUCTION

Cooperative vehicular applications depend on the timely wireless dissemination of safety-relevant information among nearby vehicles. Vehicle-to-Everything (V2X) communication enables vehicles to exchange messages directly with other vehicles, infrastructure, and pedestrians, and Institute of Electrical and Electronics Engineers (IEEE) 802.11 remains an important reference technology for this purpose because it supports direct, decentralized exchanges without continuous infrastructure support (JIANG; DELGROSSI, 2008; IEEE, 2021). In particular, IEEE 802.11p adapts the standard for vehicular operation, typically at 5.9 GHz with 10 MHz channelization, and relies on fully distributed channel access: there is no central scheduler that can reserve transmission opportunities for urgent traffic.

Among the messages exchanged in such networks, crash alerts occupy a special position. They are rare, but when they occur, their usefulness depends almost entirely on timely delivery to nearby vehicles. Contention-based channel access, however, exposes this urgent crash-related traffic to the same medium competition faced by routine transmissions: as offered load rises, delay and jitter become unstable, backoff freezing can postpone emergency frames while the channel is busy, and Quality of Service (QoS) markings are not handled consistently across deployments (SZIGETI; HENRY; BAKER, 2018; WU et al., 2019). A crash alert that arrives late competes for the channel exactly when the surrounding traffic is most likely to be dense, which is also when its delivery matters most.

Within IEEE 802.11, two standard mechanisms are central to traffic differentiation. At the network layer, Differentiated Services Code Point (DSCP) marking expresses traffic criticality; at the link layer, Enhanced Distributed Channel Access (EDCA) translates that criticality into differentiated medium access through access categories with distinct contention parameters (IEEE, 2021). The resulting wireless behavior, however, depends on the local policy that maps DSCP values into 802.11 user priorities and then into EDCA queues, and default mappings are ambiguous in practice (SZIGETI; HENRY; BAKER, 2018). Moreover, EDCA provides statistical preference rather than guarantees: higher-priority categories obtain access earlier on average, but all categories still contend randomly, and urgent traffic improves only at the expense of lower-priority flows.

Recent literature pursues stronger contention control through slot-based schedulers inspired by Time-Sensitive Networking (TSN), preemptive or cooperative Medium Access Control (MAC) schemes, and cellular sidelink resource allocation (FERAUDO; GARBUGLI; BELLAVISTA, 2024; LI et al., 2024; LINN; LIU; GAO, 2023; THABET; BENDARY; ELBAYOUMY, 2025). These approaches confirm the value of tighter contention control, but they rely on synchronization, infrastructure, or richer protocol ma-

chinery, assumptions that are difficult to satisfy in a decentralized vehicular ad hoc network. This leaves a practical gap for a local, event-driven design that isolates one crash-triggered high-priority flow against ordinary background traffic and quantifies both the protection obtained and the penalty imposed. Staying within standard IEEE 802.11p mechanisms keeps such a design deployable: no clock synchronization, infrastructure, or protocol changes are required, and adoption can proceed per node.

This dissertation addresses that gap with a deliberately narrow study: a focused crash-aware prioritization strategy built from standard mechanisms rather than a new MAC protocol. All vehicles exchange periodic Best Effort (BE) multicast traffic, while one designated crash vehicle emits temporary alert bursts marked with DSCP 46 and mapped to the Voice (VO) access category. On top of this two-class model, five MAC policies are compared: a plain Distributed Coordination Function (DCF) baseline, EDCA with explicit DSCP-to-Voice mapping, and three adaptive extensions (`stable`, `guarded`, and `emergency`) that temporarily suppress or drop competing BE traffic while alerts are active. The adaptive profiles are fixed design points that map a protection-versus-cost frontier, aiming at better statistical latency under congestion rather than deterministic guarantees.

1.1 RESEARCH QUESTION AND SCOPE

The central question investigated in this dissertation is:

> Can crash-triggered traffic obtain better wireless service than ordinary traffic in Veins/INET Framework (INET) simulations, and what cost does that protection impose on ordinary traffic?

The scope is intentionally minimal. The study considers a single crash source, a two-class BE/VO traffic distinction, and one-hop multicast dissemination on a highway corridor, with no access points, roadside units, or centralized coordination. It does not attempt to reproduce a full Intelligent Transportation Systems (ITS) stack, standardized cooperative-awareness message schedules, or multi-class traffic orchestration. All findings are obtained by simulation in Objective Modular Network Testbed in C++ (OMNeT++)/Veins/INET, and all observed prioritization is statistical contention shaping: no deterministic slotting, synchronization, or provable worst-case bounds are claimed.

1.2 OBJECTIVES

The general objective of this dissertation is to evaluate, through simulation, whether crash-triggered Voice traffic can obtain measurably better wireless service than ordinary Best Effort traffic in decentralized vehicular Wi-Fi, using only local, standard-

compliant MAC mechanisms, and to quantify the cost that this protection imposes on ordinary traffic.

The specific objectives are:

1. To implement a crash-triggered two-class traffic model in Veins/INET, in which all vehicles generate periodic BE multicast traffic and a designated crash vehicle emits DSCP 46 alert bursts mapped to the VO access category through an explicit classifier.
2. To design and implement a lightweight adaptive MAC controller that extends EDCA with a local three-mode protection mechanism, exposed as a ladder of five policies with increasing aggressiveness, from plain DCF to emergency preemption.
3. To evaluate the five policies across a matrix of vehicle densities (10 and 100 vehicles), offered-load profiles, and spatial concentrations of that load—from a uniform fleet-wide offer to a localized hotspot near the crash source—using class-specific delay, jitter, multicast reach, and MAC-loss metrics.
4. To characterize the resulting protection-versus-cost frontier and derive engineering guidance on when adaptive suppression should be enabled.

1.3 CONTRIBUTIONS

This dissertation makes the following primary contributions:

1. A lightweight adaptive MAC controller built solely from standard mechanisms—explicit DSCP 46-to-Voice mapping plus a local three-mode extension of EDCA—applied to a crash-triggered two-class traffic model in which ordinary vehicular multicast traffic remains Best Effort while a designated crash node injects temporary Voice alert bursts.
2. An open-source Veins/INET simulation artifact (AGUIAR; GRACIOLI, 2026) comparing a plain IEEE 802.11p DCF baseline, an EDCA baseline with explicit mapping, and three adaptive profiles (`stable`, `guarded`, `emergency`) across density and load matrices.
3. Engineering guidance on *when* adaptive suppression should be enabled, expressed as a regime-conditional protection-versus-cost frontier: under a uniform Poisson offer, Voice alert age does not separate the five MAC policies (about 2.2 ms P95), and blocking without dropping only inflates Best Effort delay (`stable`: 0.51 ms to 30 ms P95); but under concentrated near-crash contention with dense alert bursts, emergency preemption recovers Voice service (about –39% mean and –21% P95 alert age versus DCF, with 30%

fewer Voice collision losses than plain EDCA)—supporting event-triggered escalation rather than always-on suppression.

1.4 DOCUMENT STRUCTURE

The remainder of this dissertation is structured as follows. Chapter 2 provides the necessary background on IEEE 802.11p, contention-based channel access, EDCA, and the DSCP-to-Voice mapping path. Chapter 3 surveys related work on vehicular QoS and stronger contention-control alternatives. Chapter 4 formalizes the system model: the highway scenario, the two-class traffic abstraction, the crash event timeline, and the five MAC policies. Chapter 5 details the realization in OMNeT++/Veins/INET. Chapter 6 reports the evaluation results and their discussion. Chapter 7 concludes.

2 BACKGROUND

This chapter develops the technical foundation required to understand crash-aware traffic prioritization over decentralized vehicular Wi-Fi. The study evaluated later in this dissertation compares five MAC policies on a shared IEEE 802.11p channel: a non-differentiated DCF baseline, EDCA with an explicit DSCP to Voice mapping, and three adaptive extensions that temporarily suppress competing BE traffic while crash alerts are active. Those policies sit on top of a long standards path that runs from application criticality, through Internet Protocol (IP)-layer marking, through 802.11 user priorities and EDCA queues, down to a contention-based radio that offers neither reserved slots nor link-layer acknowledgements for group-addressed frames. The remainder of the dissertation presupposes that this path—and its failure modes under load—is understood in detail.

The chapter is organized as follows. Section 2.1 introduces vehicular safety communication and the latency budgets that motivate the work. Section 2.2 presents IEEE 802.11 DCF as the non-differentiated baseline. Section 2.3 develops EDCA, its statistical nature, and the later QoS amendments that still leave group-addressed traffic poorly served. Section 2.4 specializes these mechanisms to IEEE 802.11p Outside the Context of a BSS (OCB) operation and contrasts Decentralized Congestion Control (DCC) with event-triggered suppression. Section 2.5 traces the full DSCP-to-air-interface path that the study's classifier implements. Section 2.6 reviews the OMNeT++/Vehicles in Network Simulation (Veins)/INET and SUMO toolchain. Section 2.7 summarizes the three background pillars that frame the research problem.

Throughout, the chapter attributes normative force only to Standards-Track documents (notably RFCs 2474, 3246, and 8325) and treats Informational RFCs (2475, 4594, 9119) as architecture, guidelines, or operational analysis. No deterministic or worst-case latency claims are made for contention-based IEEE 802.11 access: EDCA and all five policies studied later provide statistical service only (KOSEK-SZOTT et al., 2013).

2.1 VEHICULAR SAFETY COMMUNICATION AND ITS REQUIREMENTS

Cooperative ITS rely on wireless dissemination of safety-relevant information among nearby vehicles, roadside equipment, and, in some architectures, network backends. V2X denotes this family of exchanges and subsumes Vehicle-to-Vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-pedestrian (V2P), and vehicle-to-network (V2N) patterns. Among the messages that circulate in such systems, two operational classes dominate the design of the lower layers. *Periodic awareness* traffic—for example Cooperative Awareness Messages (CAMs) in European Telecommunications Standards Institute (ETSI) ITS-G5 or Basic Safety Messages (BSMs) in

Dedicated Short-Range Communications (DSRC)—is emitted continuously by every equipped station so that neighbors can maintain a local dynamic map. *Event-driven safety* traffic—crash notifications, hard-braking warnings, and related Decentralized Environmental Notification Messages (DENMs)—is rare, short-lived, and high-impact: its usefulness collapses if delivery is postponed until after the surrounding vehicles have already reacted, or failed to react, on the basis of their own sensors.

These two classes compete for the same wireless medium. A crash alert that must reach nearby vehicles within a few tens of milliseconds does so precisely when the local channel is most likely to be loaded by periodic awareness traffic and by the ordinary background flows that this dissertation abstracts as Best Effort multicast. The dissemination primitive for both classes is one-hop broadcast or multicast rather than unicast: every vehicle in radio range must receive the same warning, and establishing per-neighbor associations or acknowledgements under high relative mobility is impractical (WU et al., 2019). As discussed in Section 2.5.4, however, group-addressed IEEE 802 frames inherit a hostile link-layer contract—no acknowledgement, no retransmission, and typically a low mandatory data rate—so the burden of timely delivery falls almost entirely on contention control and on the correctness of the priority mapping that admits urgent frames into the highest access category (PERKINS et al., 2021).

2.1.1 Quantified latency and reliability budgets

Service requirements for V2X scenarios are stated quantitatively by 3GPP (2020) in TS 22.186. Table 5.2-1 of that specification lists performance targets for vehicle platooning as a function of the degree of automation. Requirement [R.5.2-004] specifies, for the lowest degree of automation, an end-to-end latency of 25 ms, a reliability of 90%, a message size of 300–400 bytes, and a rate of 30 messages per second. Harsher rows in the same table raise reliability and tighten latency as automation increases; the 25 ms figure is therefore a conservative, widely cited reference bound rather than the most stringent requirement in the document. The broader 5G service-requirements specification of 3GPP (2025) (TS 22.261) defines Ultra-Reliable Low-Latency Communication (URLLC) classes in general terms but defers the concrete V2X Key Performance Indicators (KPIs) to TS 22.186; citing the latter for platooning and related scenarios is therefore the accurate practice.

These budgets are application-layer end-to-end targets. They do not imply that a MAC-layer study must meet them, nor that a sub-millisecond MAC delay constitutes a safety outcome. They do, however, establish the order of magnitude against which MAC-level tail shaping is later interpreted: when absolute VO delays remain far below 25 ms, the evaluation claims headroom under synthetic saturation rather than a proven crash-mitigation benefit; when concentrated contention pushes baseline delays toward that bound, the claim becomes tail recovery against it. That framing is returned to in

the evaluation discussion; the present chapter needs only the budget itself.

2.1.2 IEEE 802.11p as the target access technology

IEEE 802.11p adapts IEEE 802.11 for vehicular operation in the 5.9 GHz band with 10 MHz channelization and introduces OCB mode, in which stations exchange frames without association, authentication, or a Basic Service Set (BSS) context (JIANG; DELGROSSI, 2008; IEEE, 2021). The European ITS-G5 access layer builds on the same physical and MAC foundations for operation in the 5 GHz ITS band (ETSI, 2020). The decisive property for this dissertation is *decentralized access*: there is no access point, roadside unit, or cellular scheduler that can reserve transmission opportunities for urgent traffic. Every station contends for a shared channel under the rules of DCF or EDCA, and any prioritization of crash alerts must therefore be realized locally, from mechanisms that a node can apply without clock synchronization or infrastructure support. Chapter 3 surveys stronger alternatives—TSN-inspired slotting, cooperative Time Division Multiple Access (TDMA), preemptive MAC designs, and cellular sidelink allocation—that relax this constraint; the background that follows explains why, within the constraint, EDCA alone is necessary but not sufficient.

2.2 IEEE 802.11 DISTRIBUTED CHANNEL ACCESS

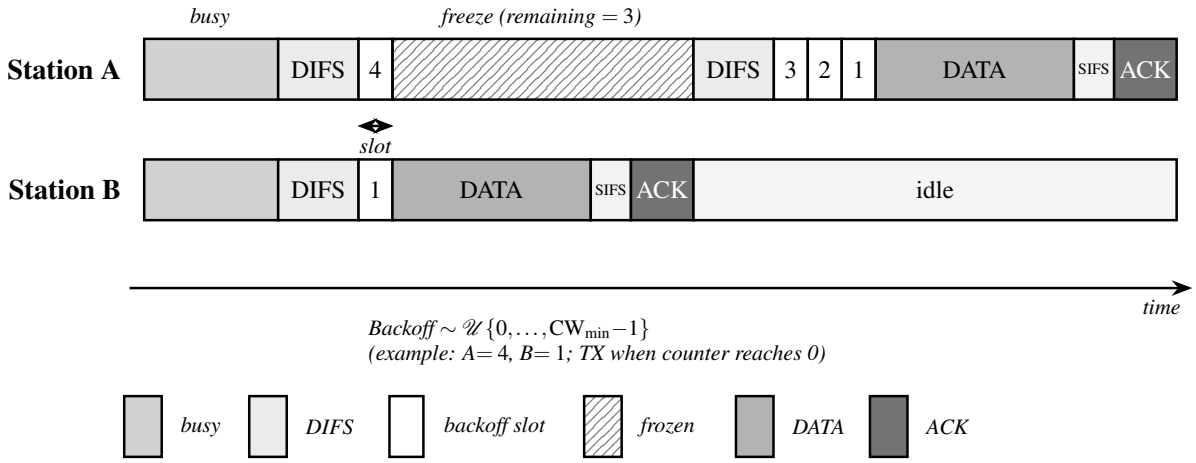
The DCF is the fundamental contention-based access method of IEEE 802.11 (IEEE, 2021). It is also the non-differentiated baseline of this study: the `plain` policy runs a single pending queue with no DSCP-to-access-category classifier and no EDCA parameter differentiation. Understanding DCF in detail is therefore prerequisite both to reading the evaluation baselines and to appreciating what EDCA adds—and what it still fails to guarantee.

2.2.1 Carrier Sense Multiple Access with Collision Avoidance

DCF implements Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) with a Binary Exponential Backoff (BEB) (BIANCHI, 2000; IEEE, 2021). A station that has a frame to transmit first senses the medium. If the channel has been idle for a DCF Interframe Space (DIFS), the station may transmit immediately. If the channel is busy, or becomes busy during the DIFS, the station defers: it draws a random backoff counter uniformly from the integer interval $[0, W - 1]$, where W is the current Contention Window (CW), and begins counting down only while the medium is sensed idle. The countdown is *frozen* whenever a transmission is detected on the channel and is resumed only after the channel has again been idle for a DIFS (BIANCHI, 2000). When the counter reaches zero, the station transmits. For unicast frames, a positive Acknowledgment (ACK) confirms success; the absence of an ACK within an ACK Timeout

triggers a retransmission after a new backoff drawn from a doubled contention window, up to a configured maximum. The contention window starts at CW_{min} and doubles after each unsuccessful transmission until it reaches CW_{max}; after a successful transmission, or after the retry limit is exhausted, it is reset. Figure 1 illustrates a two-station contention episode, including a freeze interval caused by an overlapping transmission.

Figure 1 – Timeline of the IEEE 802.11 DCF backoff process, including carrier sensing, DIFS deferral, slotted countdown, and backoff freezing while the medium is busy.



Source: Prepared by the author.

Two properties of this process are central to the later argument. First, *backoff freezing is not an anomaly*: it is part of baseline DCF and becomes the dominant contribution to access delay when the channel is repeatedly busy (BIANCHI, 2000; WU et al., 2019). Second, every station that runs a single DCF queue with the same CW_{min} and CW_{max} has the same medium-access priority, regardless of the criticality of its traffic (MANGOLD et al., 2003; SZIGETI; HENRY; BAKER, 2018). DCF therefore provides no QoS differentiation whatsoever.

2.2.2 Saturation throughput and the cost of contention

Bianchi's classical analysis models the per-station backoff process under saturation—every station always has a packet ready to send—as a two-dimensional discrete-time Markov chain whose state is the pair (backoff stage, backoff counter), as developed by Bianchi (2000). Under the approximation that each transmission attempt collides with a constant conditional probability p independent of retransmission history, the chain yields a nonlinear system in p and in the per-slot transmission probability τ, with a unique solution. From τ and p one obtains the saturation throughput of the Basic Access and Request to Send (RTS)/Clear to Send (CTS) modes.

The qualitative consequences for this dissertation do not require reproducing the full derivation. Three results matter. First, saturation throughput is the maximum load the system can carry in stable conditions; beyond that point queues grow and delay becomes unbounded in the offered-load sense (BIANCHI, 2000). Second, under Basic Access—the mode relevant to short safety frames and to group-addressed traffic that cannot use RTS/CTS—throughput degrades as the number of contending stations grows, because collisions waste large channel intervals on full-length frames (BIANCHI, 2000). Third, the contention-window parameters that maximize throughput depend on the network size, yet CW_{min} and the maximum backoff stage m are fixed by the standard; in some regimes the resulting throughput is therefore "significantly lower than the maximum achievable," and adaptive tuning of W and m is a natural response (BIANCHI, 2000). That observation is the earliest conceptual hook for the adaptive MAC profiles evaluated later: when fixed contention parameters cannot track the operating point, local, load- or event-driven adaptation becomes an engineering alternative.

2.2.3 Implications for the `plain` baseline

Taken together, Sections 2.2.1–2.2.2 define the `plain` policy of this study. A single DCF pending queue, no classifier, and identical contention parameters for every frame imply that crash-alert and background packets receive identical MAC treatment. Any difference that later appears between their measured delay statistics under `plain` must therefore be attributed to differences in the traffic processes themselves—number of senders, activity window, payload size—and not to prioritization. That sampling caveat is stated explicitly in the evaluation discussion; the background needed to understand it is the absence of differentiation in DCF (MANGOLD et al., 2003; SZIGETI; HENRY; BAKER, 2018).

2.3 QUALITY OF SERVICE IN IEEE 802.11: EDCA AND ITS LIMITS

IEEE 802.11e introduced the Hybrid Coordination Function (HCF) as the QoS-aware MAC coordination function, combining contention-based EDCA with the polling-based HCF Controlled Channel Access (HCCA) (IEEE, 2005; MANGOLD et al., 2003; IEEE, 2021). HCCA can bound delay under a single Hybrid Coordinator by granting Transmission Opportunity (TXOP) according to a schedule, but it requires a coordinator, does not solve inter-cell interference without additional coordination, and is not available in fully decentralized OCB operation (MANGOLD et al., 2003; KOSEK-SZOTT et al., 2013). This dissertation therefore focuses exclusively on EDCA, which is the mechanism underlying the `edca_only` policy and the three adaptive extensions.

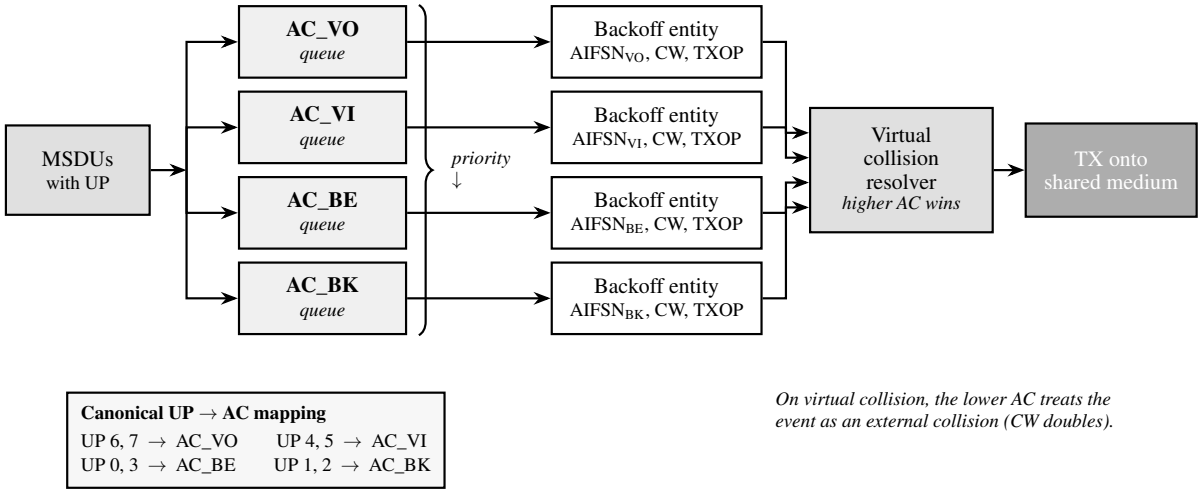
2.3.1 EDCA access categories and contention parameters

EDCA maintains four Access Categories (ACs)—Voice (`AC_VO`), Video (`AC_VI`), Best Effort (`AC_BE`), and Background (`AC_BK`)—as parallel backoff entities within a station (IEEE, 2005; MANGOLD et al., 2003; KONG et al., 2004). Each category has its own transmit queue and its own set of contention parameters: an Arbitration Interframe Space Number (AIFSN), a minimum and maximum contention window (CW_{min}[AC], CW_{max}[AC]), and an optional TXOP limit. The Arbitration Interframe Space (AIFS) of category *i* is built from the Short Interframe Space (SIFS) as

$$\text{AIFS}[i] = \text{SIFS} + \text{AIFSN}[i] \cdot a\text{SlotTime}, \tag{1}$$

with AIFSN[*i*] ≥ 2 in the standardized parameter sets (MANGOLD et al., 2003; SZIGETI; HENRY; BAKER, 2018). A smaller AIFSN, a smaller CW_{min}, and a smaller CW_{max} all raise the probability that the corresponding category wins a contention episode. When two local categories expire their backoff counters in the same slot, an *internal* (virtual) collision is resolved in favor of the higher-priority category; the lower-priority category behaves as if an external collision had occurred and doubles its contention window (KONG et al., 2004; MANGOLD et al., 2003). Figure 2 summarizes the four-queue architecture and the mapping from IEEE 802.1D user priorities into access categories.

Figure 2 – EDCA architecture: four parallel access-category queues with independent backoff entities, virtual-collision resolution, and the canonical user-priority to access-category mapping.



Source: Prepared by the author.

Table 1 reports representative EDCA parameter relationships as summarized by Request for Comments (RFC) 8325 for infrastructure WLANs; the numerical defaults

used in the present study's INET configuration are stated in the Implementation chapter (Table 12) and need not coincide with every published draft-era table (SZIGETI; HENRY; BAKER, 2018). The essential ordering is stable across revisions: `AC_VO` is the most aggressive category, followed by `AC_VI`, `AC_BE`, and `AC_BK`.

Table 1 – Representative EDCA differentiation relative to Best Effort, following the UP→AC and parameter discussion in RFC 8325. Absolute defaults vary by band and standard revision; the study's configured values are given in Table 12.

Access category	Typical UP	Relative AIFSN	Relative CW_{min}	Relative CW_{max}
`AC_VO`	6, 7	shortest	smallest	smallest
`AC_VI`	4, 5	short	small	small
`AC_BE`	0, 3	medium	large	large
`AC_BK`	1, 2	longest	large	large

Source: Adapted from Szigeti, Henry, and Baker (2018).

Kong et al. provide an analytical model of EDCA that accounts for virtual collisions, heterogeneous AIFS values, and heterogeneous contention windows, and that yields throughput and mean access delay per category (KONG et al., 2004). The model confirms that service differentiation is effective under moderate load but does not convert random access into a reservation service: every category still draws a random backoff, and an ongoing transmission from any station occupies the medium for all others.

2.3.2 Statistical priority, not guarantees

The literature is unambiguous that EDCA provides relative, statistical differentiation rather than absolute delay or throughput bounds. Mangold et al. show by simulation that higher-priority access categories restrain the throughput of lower-priority ones as offered load increases, yet that EDCA delays themselves "increase unpredictably" with load, whereas HCCA delays remain below a TXOP-defined threshold under a single coordinator (MANGOLD et al., 2003). Kosek-Szott et al. state the same conclusion directly: "Using only these parameters, EDCA cannot guarantee any throughput or delay bounds, but only performance differentiation among the categories" (KOSEK-SZOTT et al., 2013). Malik et al. further observe that MAC-layer differentiation "can ensure QoS only when network load is reasonable"; beyond a load limit, even high-priority traffic loses its advantage, and that a strict priority discipline can starve lower-priority flows to a "zero-valued throughput" under a sustained high-priority arrival process (MALIK et al., 2015).

These facts have two consequences for the present work. First, mapping crash alerts into `AC_VO` is expected to improve their average and tail delay relative to Best

Effort, but the improvement remains contingent on the contention regime and cannot be claimed as a bound. Second, the starvation of lower-priority traffic under aggressive high-priority service is not an exotic failure mode: it is the ordinary cost of prioritization. The adaptive profiles evaluated later deliberately impose a *bounded, event-triggered* form of that cost—deferring or dropping BE while alerts are active—rather than leaving suppression to the uncontrolled side-effect of sustained VO load. Framing the cost as intentional and measurable is therefore consistent with the QoS literature, not a departure from it (MALIK et al., 2015; KOSEK-SZOTT et al., 2013).

2.3.3 Adaptive EDCA and the design space for local control

Because fixed EDCA parameters cannot track every operating point, a substantial body of work retunes contention parameters in response to observed traffic. Romdhani, Ni, and Turletti proposed Adaptive Enhanced Distributed Coordination Function (EDCF), which adjusts persistence factors based on estimated collision rates to improve differentiation under load (ROMDHANI; NI; TURLETTI, 2003). Surveys classify such schemes alongside distributed fair scheduling, varying interframe spaces, differentiated frame lengths, admission control, and related MAC enhancements (NI; ROMDHANI; TURLETTI, 2004; MALIK et al., 2015). The common theme is continuous, measurement-driven adaptation of the contention process itself.

The mechanism studied in this dissertation belongs to a neighboring but distinct corner of the design space. It does not retune AIFSN or CW values. It leaves the EDCA parameter set of `edca_only` unchanged and instead interposes a local three-mode controller that temporarily withholds BE channel-access requests—and, in the emergency profile, may drop BE—while crash-related Voice activity is detected. The adaptation is therefore *event-triggered and per-class*, not a permanent retuning of aggregate contention parameters. Related Work develops the contrast with adaptive EDCA, with DCC, and with synchronized or infrastructure-assisted designs; the background point is simply that local, load-aware control of who is allowed to contend is a recognized response to the limits of static EDCA (ROMDHANI; NI; TURLETTI, 2003; MALIK et al., 2015).

2.3.4 Later QoS amendments and the group-addressed gap

IEEE 802.11aa and IEEE 802.11ae later extended the QoS toolkit for infrastructure WLANs (KOSEK-SZOTT et al., 2013). 802.11aa introduced the Group Addressed Transmission Service (GATS), including Groupcast with Retries (GCR) Unsolicited Retry and GCR Block Ack, to raise the delivery probability of group-addressed frames that the base standard transmits without retries. It also added intra-AC queue splitting for Voice and Video and a Stream Classification Service (SCS) that can map streams to queues independently of the 802.1D user-priority table. 802.11ae introduced the

QoS Management Frame (QMF) policy so that management frames need not always consume the highest access category.

These amendments are evidence of an acknowledged gap: standard group-addressed delivery was recognized as unreliable or unscalable, and substantial protocol machinery was added to repair it in BSS operation (KOSEK-SZOTT et al., 2013). They do not, however, transfer to the OCB setting of this study. GCR, directed multicast, and related mechanisms centred on an Access Point (AP) presuppose an infrastructure BSS, association state, and often a coordinator; none of these are present when vehicles exchange frames Outside the Context of a BSS. The crash-alert multicast of this dissertation therefore inherits the unresolved group-addressed contract of the base standard, as elaborated in Section 2.5.4.

2.4 IEEE 802.11P AND VEHICULAR AD HOC OPERATION

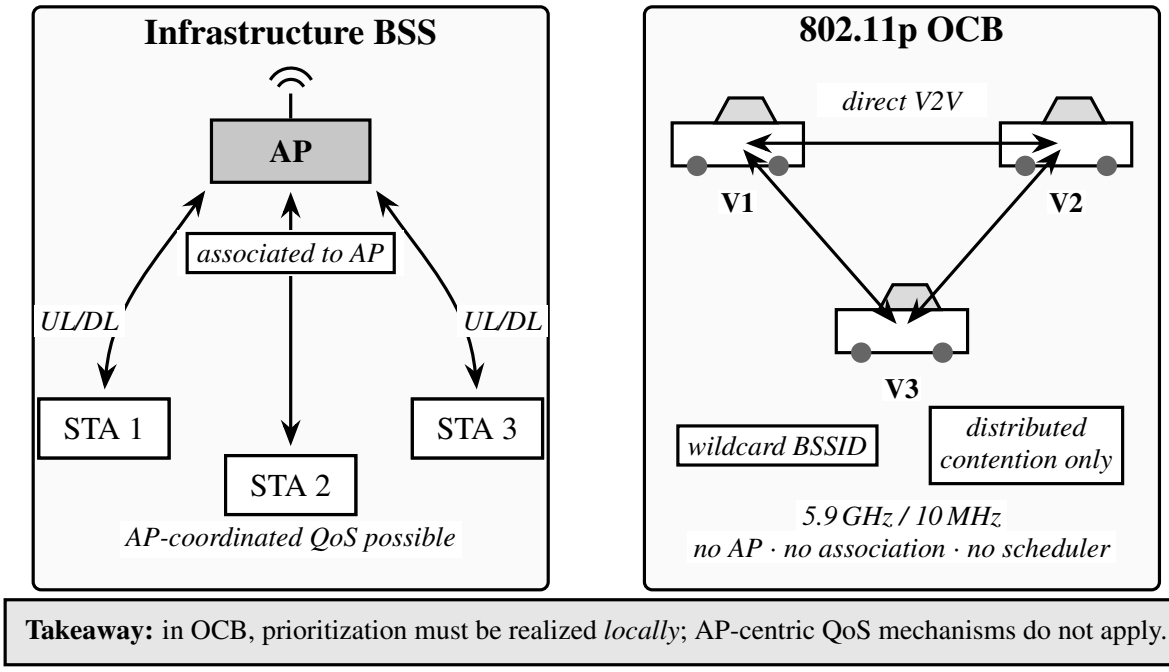
The mechanisms of Sections 2.2–2.3 are realized, in this work, on an IEEE 802.11p-like radio operating in OCB mode. This section specializes the background to that environment and introduces DCC as the standardized, always-on counterpart to the study's event-triggered suppression.

2.4.1 OCB mode, channelization, and multi-channel context

IEEE 802.11p was designed so that vehicles can exchange frames during brief contact intervals without the overhead of joining a BSS (JIANG; DELGROSSI, 2008). In OCB mode, frames carry a wildcard BSSID; stations neither associate nor authenticate with an access point; and management procedures that assume a BSS are omitted (IEEE, 2021). The physical layer typically uses 10 MHz Orthogonal Frequency-Division Multiplexing (OFDM) channels in the 5.9 GHz band, doubling the symbol duration relative to 20 MHz 802.11a and improving robustness to multipath and Doppler at highway speeds (JIANG; DELGROSSI, 2008; ETSI, 2020). Figure 3 contrasts BSS-mode infrastructure Wi-Fi with OCB vehicular operation.

Above the 802.11p MAC, the IEEE 1609.4 multi-channel specification organizes time into synchronization intervals that alternate a Control Channel interval with a Service Channel interval, so that safety and non-safety traffic can be separated in frequency and time (IEEE, 2016). The present study does not model that multi-channel schedule. It evaluates MAC policies on a single shared channel under controlled offered load, holding the physical layer fixed across configurations so that BE–VO differences isolate queueing and contention effects. IEEE 1609.4 remains relevant context—and appears in related work on multi-channel scheduling (GARRIDO ABENZA et al., 2024)—but is outside the experimental scope.

Figure 3 – Comparison of infrastructure BSS operation and IEEE 802.11p Outside the Context of a BSS (OCB) vehicular operation. Prioritization in OCB must be realized locally; AP-centric QoS mechanisms do not apply.



Takeaway: in OCB, prioritization must be realized *locally*; AP-centric QoS mechanisms do not apply.

Source: Prepared by the author.

2.4.2 EDCA under OCB and the broadcast safety contract

EDCA is available in vehicular deployments and is the mechanism by which a correctly mapped crash alert obtains `AC_VO` treatment. Its limitations, however, interact poorly with the broadcast safety contract. Group-addressed frames are transmitted without link-layer acknowledgement and without retransmission; the sender cannot detect loss and therefore does not apply collision-driven backoff on behalf of failed receivers (KOSEK-SZOTT et al., 2013; PERKINS et al., 2021). Priority markings still govern which local queue is served and how aggressively that queue contends, but they cannot recover a frame that collided or faded at a particular neighbor. In a dense highway scenario, the same density that raises the value of a crash warning also raises the collision probability and the frequency of backoff freezing, so the frames that most need prompt channel access are deferred by the very congestion they are meant to mitigate (WU et al., 2019).

Wu et al. analyze IEEE 802.11p broadcast performance under continuous backoff freezing and show that accounting for repeated freeze intervals yields higher access delay than models that ignore the effect, and that both delay and packet delivery ratio degrade as vehicle density increases (WU et al., 2019). Their absolute numerical delays correspond to a lightly loaded configuration and are not transferable to the saturated BE

loads of this study; the *trend*—freezing inflates delay under load, and density worsens both delay and delivery—is the transferable claim and matches the qualitative prediction of Bianchi's saturation analysis (BIANCHI, 2000).

2.4.3 Decentralized Congestion Control

European ITS-G5 deployments complement EDCA with DCC at the access layer (ETSI, 2018). DCC measures the Channel Busy Ratio (CBR)—the fraction of time the medium is sensed busy—and applies Transmit Power Control (TPC), Transmit Rate Control (TRC), and DCC access control so that each station's contribution to channel load remains within a target operating region. Both reactive and adaptive algorithm variants are defined; in all cases the control loop acts *continuously* on a station's *aggregate* offered load, across traffic classes, as a function of local (and optionally shared) CBR (ETSI, 2018).

DCC and the adaptive profiles of this dissertation therefore address related but distinct problems. DCC is a permanent congestion-management regime that throttles how much traffic a station injects as the channel fills. The crash-aware controller intervenes only while a crash alert is active, and it discriminates between BE and VO rather than scaling all traffic uniformly. The two mechanisms are complementary: DCC can keep the long-term operating point away from collapse, while event-triggered BE suppression can temporarily reshape contention during a short alert window. The evaluation deliberately does not enable DCC, so that the protection-versus-cost frontier of the five MAC policies is isolated; claiming superiority over DCC is neither intended nor supported.

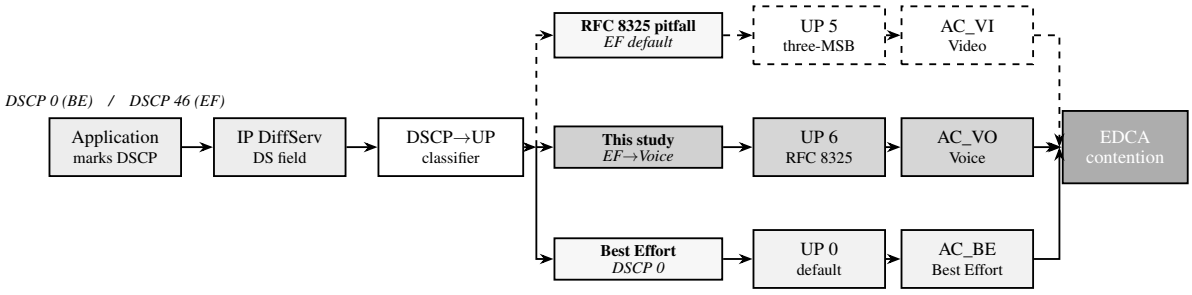
2.5 FROM DSCP TO THE AIR INTERFACE

Application criticality reaches the wireless MAC only through an explicit cross-layer mapping when IP-layer marking is used. This section traces that mapping from the Differentiated Services (DiffServ), through the Expedited Forwarding (EF) per-hop behavior and DSCP 46, through the DSCP-to-user-priority-to-access-category path standardized for Wi-Fi, to the multicast limitations that still apply after a correct mapping. The path is exactly what the study's `QosClassifier` implements for crash alerts and is the reason the dissertation carefully distinguishes DSCP 46, the EF per-hop behavior, user priority 6, and `AC_VO`: they are related objects on a chain, not synonyms. Figure 4 summarizes that cross-layer path and the RFC 8325 Voice-mapping pitfall.

2.5.1 Differentiated Services architecture

RFC 2474 defines the Differentiated Services (DS) field in Internet Protocol version 4 (IPv4) and Internet Protocol version 6 (IPv6) headers: six bits form the

Figure 4 – Cross-layer path from IP DSCP marking to EDCA channel access. Correct Voice treatment of Expedited Forwarding traffic requires an explicit DSCP-to-user-priority mapping; the default three-MSB heuristic places EF in Video.



Source: Prepared by the author.

DSCP used to select a Per-Hop Behavior (PHB) at each node, and DS-compliant nodes must select PHBs by matching the entire six-bit field against a configurable mapping (NICHOLS et al., 1998). Packets received with an unrecognized codepoint should be forwarded as if marked for the Default behavior; the Default PHB is ordinary best-effort forwarding, and its recommended codepoint is `000000` (DSCP 0) (NICHOLS et al., 1998). RFC 2475, an Informational architecture document, situates these codepoints in a scalable edge-core model: complex classification and conditioning occur at network boundaries or hosts, while interior nodes apply PHBs to behavior aggregates (BLAKE et al., 1998). A Behavior Aggregate (BA) classifier selects packets on the DSCP alone; Multi-Field (MF) classifiers may additionally inspect addresses, ports, or other header fields. Traffic sources within a domain may pre-mark packets so that an application's preferences are reflected before the first hop (BLAKE et al., 1998).

Two architectural observations are directly relevant. First, a DSCP mark confers nothing unless some node maps it to a PHB—and, on a shared wireless link, to a link-layer service (BLAKE et al., 1998). RFC 2475 already notes that extending DiffServ semantics across Layer 2 infrastructures depends on "the mapping of PHBs to different link-layer services," and that under a restricted set of priority classes PHBs may be supportable only partially or may become indistinguishable. Second, host pre-marking is a first-class part of the architecture: the crash vehicle that marks its alerts with DSCP 46 is exercising the same pattern that RFC 2475 describes for sources that know their application's requirements.

2.5.2 Expedited Forwarding and DSCP 46

RFC 3246 defines the EF PHB as a building block for low-delay, low-jitter, and low-loss services, obtained by ensuring that the EF aggregate is served at a configured rate independently of the non-EF load (DAVIE et al., 2002). Codepoint `101110`—decimal 46—is recommended for the EF PHB (DAVIE et al., 2002). The PHB specification is per-node; constructing an end-to-end service from EF hops is out of scope of the RFC and, on a single shared wireless hop, is not the claim this dissertation makes.

RFC 4594 provides Informational configuration guidelines that map application service classes onto DiffServ PHBs (BABIARZ; CHAN; BAKER, 2006). The Telephony service class is recommended for inelastic, real-time, very-low-delay, very-low-jitter, and very-low-loss traffic; it should use the EF PHB, should be configured with priority queuing, and should be pre-marked with the EF DSCP at the application or end point (BABIARZ; CHAN; BAKER, 2006). The Standard service class must use the Default Forwarding PHB with DSCP 0. Crash alerts are not IP telephony. They do, however, match the *traffic characteristics and required performance* on which RFC 4594 bases class definitions: inelastic short bursts that tolerate little delay or loss. Marking them with DSCP 46 and admitting them to the most aggressive MAC category is therefore the DiffServ-consistent modeling choice for the two-class abstraction, not a claim that production ITS stacks universally use EF for DENM-like messages.

2.5.3 Mapping DiffServ to IEEE 802.11

RFC 8325 is the Standards-Track reference for mapping DiffServ to IEEE 802.11 (SZIGETI; HENRY; BAKER, 2018). Its central operational finding is that *no standard governs the default DSCP-to-user-priority mapping*: IEEE 802.11 does not define how upper-layer markings should map to user priorities, and the example tables in the standard are not normative recommendations (SZIGETI; HENRY; BAKER, 2018). Industry practice often uses the three most significant bits of the DSCP as the user priority. Under that heuristic, EF (`101110`) maps to user priority 5 and is therefore admitted to `AC_VI` (Video) rather than to `AC_VO` (Voice), “for which it is intended” (SZIGETI; HENRY; BAKER, 2018). RFC 8325 consequently recommends a non-default mapping in which EF is mapped to user priority 6 and thence to `AC_VO`.

User-priority markings alone do not affect over-the-air treatment; only the subsequent mapping of user priorities into access categories, and the contention parameters of those categories, produce differentiation (SZIGETI; HENRY; BAKER, 2018). Table 2 summarizes the default pitfall and the recommended Voice mapping for the two codepoints used in this study.

RFC 8325 also states an honesty clause that this dissertation adopts verbatim in spirit: IEEE 802.11 “does not support an EF PHB service, as it is not possible to assure

Table 2 – DSCP to user-priority to access-category mapping for the study's two traffic classes, contrasting the common three-MSB default with the RFC 8325 recommendation for Expedited Forwarding.

Traffic	DSCP	Default 3-MSB UP	Recommended UP	Resulting AC
Background (BE)	0 (000000)	0	0	AC_BE
Crash alert (EF)	46 (101110)	5 → AC_VI	6	AC_VO

Source: Adapted from Szigeti, Henry, and Baker (2018).

that a given access category will be serviced with strict priority over another (due to the random element within the contention process)" (SZIGETI; HENRY; BAKER, 2018). Correct DSCP-to-Voice mapping is therefore necessary but not sufficient. It places crash alerts in the right queue with the right contention parameters; it does not reserve the channel, silence neighbors, or bound the delay of a Voice frame that must wait for an ongoing transmission and for backoff freezes. That residual gap is exactly why the adaptive BE-suppression profiles exist on top of `edca_only`.

A further operational caveat, developed in the evaluation discussion, follows from the same document: any client that can mark packets with preferred user priorities can monopolize airtime, creating a WLAN denial-of-service vector (SZIGETI; HENRY; BAKER, 2018). In the vehicular setting, an unauthenticated or faulty source that continuously emits VO-marked frames could keep neighbors in a suppression regime. Finite protection windows bound but do not eliminate that risk; authentication and rate limiting are deployment prerequisites, not contributions of the present MAC study.

RFC 8325 is written for infrastructure WLANs, with upstream and downstream directions relative to an access point. The DSCP→UP→AC mechanics carry over unchanged to OCB because they are properties of the local MAC stack; the AP-centric recommendations (QoS Map element distribution, directed multicast services, and similar) do not, and are not assumed here.

2.5.4 Multicast considerations over IEEE 802 wireless media

RFC 9119 analyzes multicast over IEEE 802 wireless media as an Informational operational document (PERKINS et al., 2021). Three limitations dominate the crash-alert setting. First, because there are no ACKs for multicast packets, the transmitter cannot know whether a retransmission is needed; there is consequently no loss-driven backoff on behalf of failed receivers, and packet loss rates of 5% or more are not uncommon (PERKINS et al., 2021). Second, multicast and broadcast traffic is generally transmitted at the slowest rate among connected devices—the basic rate—so that every intended receiver can decode the frame; spectral-efficiency techniques such as multi-user Multiple-Input Multiple-Output (MIMO) spatial multiplexing are unavailable with

more than one intended receiver (PERKINS et al., 2021). Third, low-rate transmissions occupy the medium longer, consuming airtime that would otherwise be available to other stations and degrading overall capacity (PERKINS et al., 2021).

Table 3 summarizes which reliability tools are available in infrastructure BSS operation versus in the OCB setting of this study. Directed Multicast Service, GCR, and related AP-based mitigations can raise delivery probability in a BSS but "still [do] not guarantee success" and may introduce unacceptable latency (PERKINS et al., 2021; KOSEK-SZOTT et al., 2013); more importantly, they are unavailable without an AP. The study therefore keeps one-hop multicast—the natural safety primitive—and concentrates engineering effort on contention shaping rather than on link-layer retransmission.

Table 3 – Link-layer reliability mechanisms for group-addressed frames: availability in infrastructure BSS Wi-Fi versus in IEEE 802.11p OCB operation as used in this study.

Mechanism	Infrastructure BSS	OCB (this study)
Link-layer ACK / retransmission for group-addressed frames	No (base standard)	No
GCR Unsolicited Retry / Block Ack (802.11aa)	Yes (if implemented)	No
Directed Multicast / multicast-to-unicast conversion	Yes (AP)	No
EDCA priority via UP→AC mapping	Yes	Yes
DCC aggregate load control (ITS-G5)	N/A	Possible (not enabled here)
Event-triggered BE suppression (this work)	N/A	Evaluated

Source: Adapted from Perkins et al. (2021).

As with RFC 8325, RFC 9119 is written around AP/BSS terminology. The Layer 2 facts required here—no ACK, no retransmission, basic-rate transmission, airtime cost—apply equally to OCB broadcast; the AP-based remedies surveyed in its later sections do not.

2.6 SIMULATION OF VEHICULAR NETWORKS

Empirical measurement of dense, instrumented highway fleets under controlled multi-policy MAC configurations is rarely feasible at research scale. Discrete-event simulation is therefore the dominant methodology for Inter-Vehicle Communication (IVC) protocol evaluation, provided that both the network stack and the mobility process are represented at adequate fidelity (SOMMER; GERMAN; DRESSLER, 2011).

2.6.1 OMNeT++ and INET

OMNeT++ is a modular, component-based C++ simulation framework and discrete-event engine widely used for communication-network research (VARGA; HORNIG, 2008). The INET Framework supplies Internet protocol stacks, link-layer models including IEEE 802.11, radio propagation abstractions, and mobility bases on top of OMNeT++ (VARGA et al., 2024). In this dissertation, INET provides the IEEE 802.11p-capable MAC and Physical Layer (PHY) baseline, IP forwarding, and the hooks into which the study's traffic generators, DSCP classifier, and adaptive MAC controller are inserted. The Implementation chapter details those modules; the background point is that the policies under test replace or configure narrow parts of an otherwise standard stack, so that cross-policy comparisons share the same PHY, mobility, and load matrix.

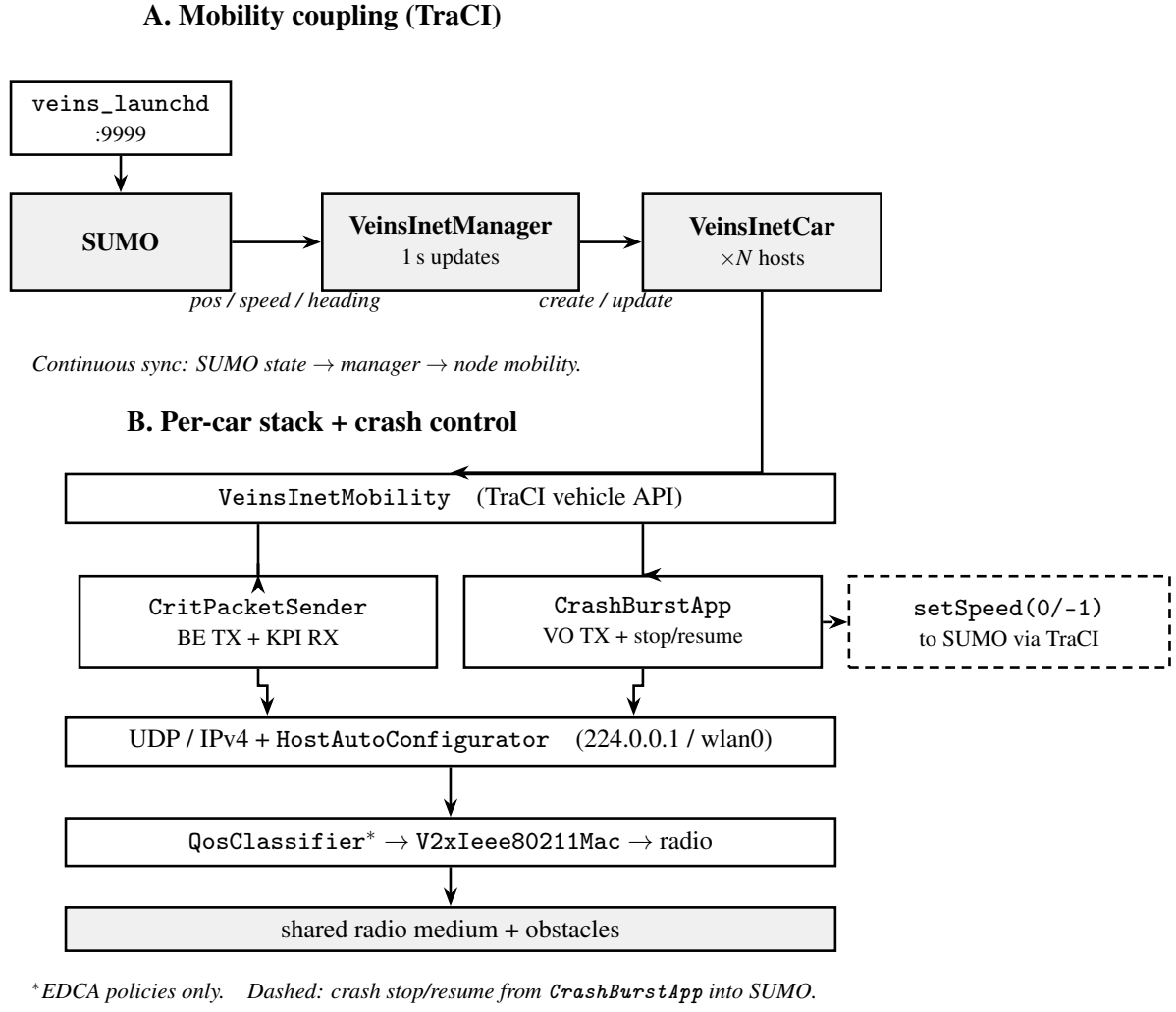
2.6.2 Bidirectional coupling with SUMO via Veins

Road-traffic mobility and network protocol behavior are not independent: a crash that stops a vehicle changes both the wireless contention neighborhood and the macroscopic traffic pattern, and a protocol that disseminates warnings can in principle alter routing decisions. Sommer, German, and Dressler argue that faithful IVC evaluation therefore requires *bidirectional* coupling of a network simulator and a road-traffic microsimulator, and introduce Veins as a hybrid framework that joins OMNeT++ to the Simulation of Urban MObility (SUMO) package through the Traffic Control Interface (TraCI) (SOMMER; GERMAN; DRESSLER, 2011). It allows the network model to control vehicle entities (for example, stopping a designated crash node) while SUMO continuously updates positions, headings, and neighborhood relations that feed the radio model.

SUMO itself is an open-source microscopic traffic simulator supporting multimodal traffic, lane-changing and car-following models, and external coupling interfaces; Alvarez Lopez et al. provide the reference publication describing its architecture and validation role in the ITS research ecosystem (LOPEZ et al., 2018). Figure 5 sketches the closed loop used in this work: SUMO advances vehicle mobility; TraCI exposes state and control; Veins/INET execute the 802.11p stack and the study's QoS modules; and KPIs are recorded from packet timestamps and MAC counters.

The present experiments use this toolchain on a highway corridor with light (10 vehicles) and heavy (100 vehicles) density packages. Physical-layer assumptions—free-space path loss, binary obstacle loss, fixed Signal-to-Noise-plus-Interference Ratio (SNIR) threshold—are held constant across MAC policies so that measured differences isolate contention and queueing. Those assumptions, and their limitations relative to fading and shadowing models, are stated with the System Model; they are background

Figure 5 – Simulation coupling as implemented in the artifact: SUMO is launched through `veins_launchd`; `VeinsInetManager` synchronizes mobility over TraCI and instantiates `VeinsInetCar` nodes; each car runs `CritPacketSender` and `CrashBurstApp` over an INET 802.11p stack. The dashed path is the crash stop/resume command from `CrashBurstApp` into SUMO via the vehicle TraCI interface.



Source: Prepared by the author.

here only insofar as they explain why a coupled microsimulation, rather than a synthetic random-waypoint model, is the appropriate mobility substrate (SOMMER; GERMAN; DRESSLER, 2011).

2.7 SUMMARY

Three pillars emerge from the preceding sections and frame the research problem addressed in the remainder of this dissertation.

1. **Safety alerts have millisecond-scale budgets but ride on contention-based access.** Cooperative V2X warnings are useful only if delivered promptly to nearby vehicles; 3rd Generation Partnership Project (3GPP) TS 22.186 places a representative platooning bound at 25 ms end-to-end (3GPP, 2020). On IEEE 802.11p in OCB mode, those warnings share a decentralized channel with no central scheduler (JIANG; DELGROSSI, 2008; IEEE, 2021). DCF treats every frame alike; under load, binary exponential backoff and repeated freezing postpone transmissions independently of criticality (BIANCHI, 2000; WU et al., 2019).
2. **EDCA provides statistical priority, and correct DSCP-to-AC mapping is not automatic.** Mapping crash alerts into `AC_VO` improves their expected access relative to Best Effort, but EDCA still contends randomly and cannot guarantee delay or throughput bounds (KOSEK-SZOTT et al., 2013; MANGOLD et al., 2003; SZIGETI; HENRY; BAKER, 2018). At the IP layer, DSCP 46 is the recommended EF codepoint for inelastic low-delay traffic (DAVIE et al., 2002; BABIARZ; CHAN; BAKER, 2006), yet the common three-MSB DSCP-to-UP heuristic places EF in Video rather than Voice; an explicit mapping to user priority 6 is required (SZIGETI; HENRY; BAKER, 2018). Even after that mapping, 802.11 cannot implement a true EF per-hop behavior on a shared medium (SZIGETI; HENRY; BAKER, 2018).
3. **Multicast alerts obtain no link-layer reliability, and standardized remedies either do not apply to OCB or act on aggregate load.** Group-addressed frames receive no ACK and no retransmission, are often sent at the basic rate, and consume disproportionate airtime (PERKINS et al., 2021). Infrastructure repairs such as GCR presuppose a BSS (KOSEK-SZOTT et al., 2013); DCC regulates each station's aggregate offered load continuously rather than suppressing BE only during an alert (ETSI, 2018). A local, event-triggered, per-class intervention that leaves EDCA parameters unchanged therefore occupies a distinct and still underexplored point in the design space.

Chapter 3 surveys stronger contention-control proposals—synchronized slotting, cooperative TDMA, preemptive and UAV-assisted MACs, and cellular sidelink allocation—and positions the present event-driven approach against them. Chapter 4 then formalizes the two-class traffic model, the crash timeline, and the five MAC policies that instantiate the background developed here.

3 RELATED WORK

Chapter 2 established that EDCA improves statistical priority for Voice traffic but cannot guarantee delay or throughput bounds on a shared wireless medium (KOSEK-SZOTT et al., 2013; MANGOLD et al., 2003), that correct DSCP to access-category mapping is not automatic (SZIGETI; HENRY; BAKER, 2018), and that group-addressed IEEE 802 frames inherit a hostile link-layer contract without acknowledgement or retransmission (PERKINS et al., 2021). Those facts define a practical research neighborhood: how to protect rare, high-impact crash alerts under contention without introducing synchronization, infrastructure, or a new MAC standard.

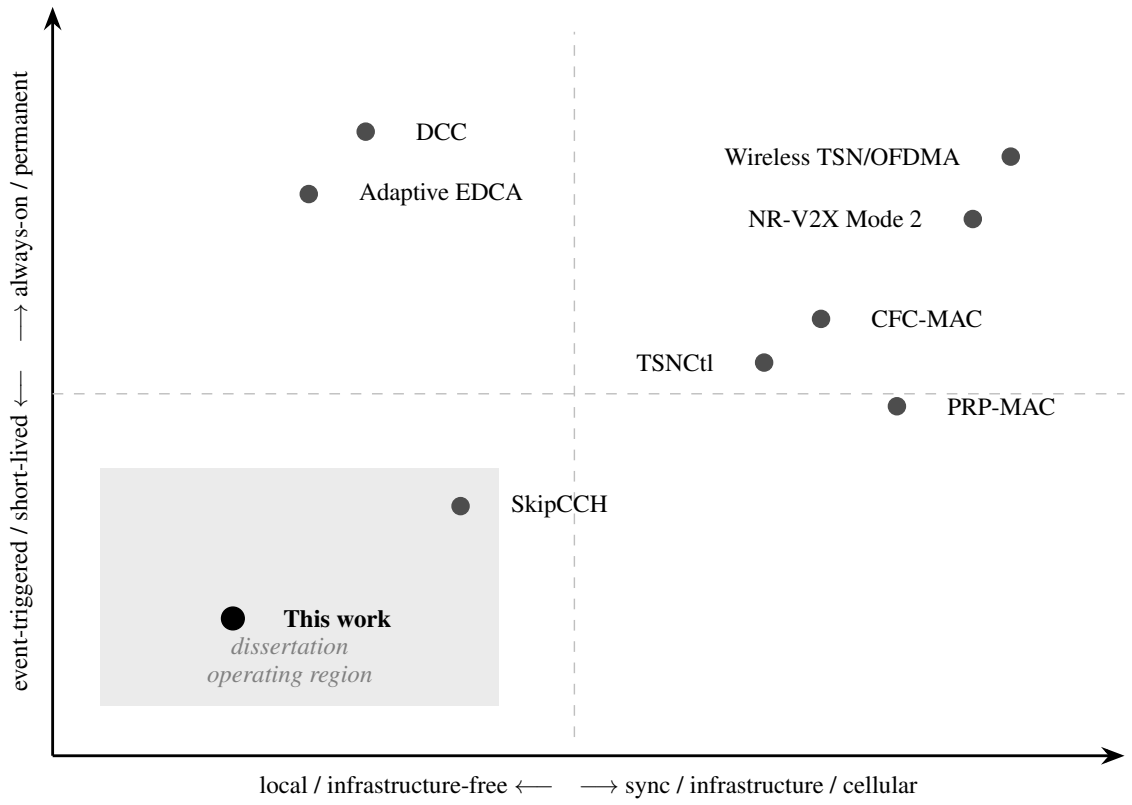
The literature answers that question along several axes. Some proposals pursue *stronger contention control* than plain EDCA through slot scheduling, preemption, cooperative TDMA, or cellular sidelink allocation. Others study *when* traffic is allowed to contend and what ordinary traffic pays when urgent flows are favored. A third family regulates offered load *continuously* through adaptive EDCA parameters or DCC. A fourth transplants TSN and Orthogonal Frequency-Division Multiple Access (OFDMA) ideas from wired or infrastructure Wi-Fi into wireless settings. Figure 6 situates these families in a design space whose axes are synchronization or infrastructure dependency and always-on versus event-triggered intervention. The figure is a schematic positioning aid: markers are placed qualitatively from each work's stated assumptions, not from a numerical score.

This dissertation occupies a deliberately narrow point in that space: explicit DSCP 46-to-Voice mapping plus local, event-triggered BE suppression on IEEE 802.11p OCB operation, without reservations, clock synchronization, or protocol changes (JIANG; DELGROSSI, 2008; IEEE, 2021). The remainder of the chapter surveys the four families above, synthesizes their assumptions and limitations, and states the research gap that Chapter 4 instantiates as a five-policy evaluation ladder.

3.1 STRONGER CONTENTION CONTROL FOR VEHICULAR NETWORKS

A recurring response to EDCA's statistical limits is to replace or overlay random access with tighter coordination. The works reviewed in this section confirm that stronger control can reduce collisions and delay for safety-oriented traffic, but they do so under operating assumptions—synchronized slotting, a coordinating master or Unmanned Aerial Vehicle (UAV), TDMA-style reservations, or cellular resource selection—that are broader than the focused crash-protection problem addressed here (FERAUDO; GARBUGLI; BELLAVISTA, 2024; LI et al., 2024; LINN; LIU; GAO, 2023; THABET; BENDARY; ELBAYOUMY, 2025).

Figure 6 – Schematic design-space taxonomy of related Medium Access Control approaches. Axes contrast local / infrastructure-free operation with synchronization, infrastructure, or cellular machinery, and event-triggered / short-lived intervention with always-on / permanent regulation. Marker positions are qualitative judgments from each work's stated assumptions, not computed scores; this dissertation occupies the local, event-triggered region.



Schematic placement only: positions are qualitative judgments from each work's stated assumptions (not computed scores or distances).

Source: Prepared by the author.

3.1.1 TSN-inspired slot scheduling for platooning

Feraudo, Garbugli, and Bellavista propose TSNCtl, an application-level controller for V2V platooning that draws inspiration from TSN slotting (FERAUDO; GARBUGLI; BELLAVISTA, 2024). The controller runs a finite-state machine that manages platoon formation and assigns time slots for message dissemination, aiming to bound collisions that arise when platoon members share a Wi-Fi-like channel. In OMNeT++/INET experiments, short slot lengths (for example, 2 ms) are reported to reduce average collision rates relative to a CSMA/CA baseline under the evaluated platooning scenarios (FERAUDO; GARBUGLI; BELLAVISTA, 2024).

The design's effectiveness rests on synchronized operation and on a platoon master that allocates slots. Those assumptions are natural for a coordinated platoon

with a designated leader; they are not available in a fully decentralized highway fleet in which a single crash source must warn arbitrary neighbors without prior group formation. TSNCtl is furthermore published as an arXiv work-in-progress preprint and should be read as preliminary rather than as a peer-reviewed standard result (FERAUDO; GARBUGLI; BELLAVISTA, 2024). For the present dissertation, the relevant takeaway is motivational: slot-based suppression of contention can protect critical V2V traffic, but only by introducing synchronization and leadership that the crash-aware study deliberately avoids.

3.1.2 Preemptive-resume priority in UAV-assisted VANETs

Li et al. address BSM delivery under high vehicle density by combining IEEE 802.11 CSMA/CA with a preemptive-resume priority scheme and an arbitration device, in a setting where a UAV acts as a receiver for BSM uploads toward Roadside Unit (RSU) processing (LI et al., 2024). Higher-priority packets can preempt lower-priority ones; queuing analysis is used to size transmission intervals so that BSM delay meets driving-safety targets under the modeled load (LI et al., 2024). Simulations support the claim that preemption reduces BSM delay relative to non-preemptive baselines in the UAV-assisted topology they study.

The architecture therefore couples richer MAC machinery (preemption and arbitration) with infrastructure assistance (UAV and, in the broader VANET framing, roadside units). Crash-alert dissemination among peer vehicles on an OCB channel has neither an arbiter nor a UAV relay. The dissertation retains the intuition that aggressive prioritization of urgent traffic can compress delay, but realizes aggression as local BE deferral or drop during an alert window rather than as a preemptive-resume service discipline under external arbitration (LI et al., 2024).

3.1.3 Contention-free cooperative TDMA

Linn, Liu, and Gao propose CFC-MAC, a distributed contention-free cooperative MAC for Vehicular Ad hoc Networks (VANETs) that combines cooperative forwarding, vectorized trajectory estimation, and time-slot reallocation to resolve heterogeneous collisions under stochastic traffic (LINN; LIU; GAO, 2023). The protocol seeks deterministic access delay and low collision rates by identifying cooperative vehicles, suppressing merging collisions through relative-velocity estimation, and reclaiming or reassigning slots after collision detection (LINN; LIU; GAO, 2023). Reported simulations show substantial collision-rate and access-delay improvements over existing solutions under the authors' scenarios.

CFC-MAC's gains presuppose a TDMA-style organization of the medium and the signaling needed to estimate trajectories and renegotiate slots. That is a stronger coordination contract than EDCA on a single shared OCB channel. The present work

does not attempt contention-free operation; it accepts random access and asks how much local, temporary BE suppression can reshape the contention that remains (LINN; LIU; GAO, 2023; IEEE, 2021).

3.1.4 Cellular NR-V2X sidelink resource allocation

Thabet, Bendary, and Elbayoumy study adaptive autonomous resource allocation for V2X sidelink, with the paper framed toward 6G; the Mode 2 sensing and semi-persistent scheduling mechanisms they adapt are those of 5G New Radio V2X (NR-V2X), improving legacy Semi-Persistent Scheduling (SPS) behavior for periodic safety-message transmissions in decentralized operation (THABET; BENDARY; ELBAYOUMY, 2025). Mode 2 allows vehicles to select radio resources without a base-station scheduler, but resource selection, sensing, and SPS persistence are still cellular sidelink mechanisms defined by the 3GPP stack rather than by IEEE 802.11 EDCA (THABET; BENDARY; ELBAYOUMY, 2025).

NR-V2X is therefore a parallel access technology, not a drop-in repair for 802.11p OCB. Comparisons that claim superiority of one stack over the other require matched scenarios and are outside this dissertation's scope. The citation's role here is to acknowledge that the industry pursues tighter resource control for periodic safety traffic through cellular machinery, whereas this study remains inside the IEEE 802.11p design envelope (JIANG; DELGROSSI, 2008; THABET; BENDARY; ELBAYOUMY, 2025).

3.1.5 Authors' prior wireless TSN exploration

An earlier conference study by Aguiar and Gracioli explored wireless TSN and QoS support for V2X under a simpler IEEE 802.11 setting, with emphasis on Time-Aware Shaper schedules and synchronization rather than on decentralized crash-triggered prioritization (AGUIAR; GRACIOLI, 2025). That work is a precursor in the sense that it investigated determinism-oriented QoS for vehicular Wi-Fi; it is not the same research question. The present dissertation drops Time-Aware Shaper (TAS)-oriented scheduling, abandons synchronization as a requirement, and focuses on a shared OCB channel in which crash-alert Voice bursts compete with ordinary BE multicast under five MAC policies that require no gate-control lists (AGUIAR; GRACIOLI, 2025).

3.1.6 Synthesis: stronger control under broader assumptions

Table 4 summarizes the approaches in this section and the families reviewed later. Assumptions and mechanisms are those stated by the cited works; the final row positions the crash-aware study. Collectively, TSNCtl, PRP-MAC, CFC-MAC, and NR-V2X Mode 2 demonstrate that tighter contention or resource control can improve

safety-oriented delivery metrics, but each relies on synchronization, a coordinating entity, TDMA-style reservations, UAV or roadside assistance, or cellular sidelink machinery (FERAUDO; GARBUGLI; BELLAVISTA, 2024; LI et al., 2024; LINN; LIU; GAO, 2023; THABET; BENDARY; ELBAYOUMY, 2025). None isolates the narrow question of whether a local, standard-compliant EDCA stack—augmented only by explicit DSCP-to-Voice mapping and short-lived BE suppression—can protect a crash-triggered multicast burst and at what cost to ordinary traffic. That narrower question is the gap this dissertation fills.

3.2 CONTENTION DYNAMICS AND THE COST OF PROTECTION

Stronger MAC designs are one response to EDCA's limits. A complementary literature asks how delay grows under vehicular load and what happens to ordinary traffic when urgent flows are favored. That literature motivates evaluating not only whether crash traffic is protected, but also how much BE traffic pays for the protection (MALIK et al., 2015; KOSEK-SZOTT et al., 2013).

3.2.1 Backoff freezing under IEEE 802.11p load

Wu, Xia, Fan, and Li analyze IEEE 802.11p broadcast performance when the backoff counter freezes continuously during busy periods (WU et al., 2019). Their model couples highway density to the number of contenders in carrier-sense range and shows that accounting for repeated freeze intervals yields higher access delay than analyses that ignore the effect; delay and packet delivery ratio both degrade as density increases (WU et al., 2019). As noted in Chapter 2, their absolute delay numbers correspond to a lightly loaded configuration and are not transferable to the saturated BE loads of this study; the transferable claim is the trend.

For crash-aware design, that trend is decisive. The same density that raises the value of a timely warning also raises the frequency of freeze intervals that postpone channel access for every contending station, including the crash source (WU et al., 2019; BIANCHI, 2000). A mechanism that only remaps DSCP to Voice still contends through random backoff; a mechanism that temporarily removes BE competitors during the alert window attacks the freeze process at its source by reducing the offered load that keeps the medium busy.

3.2.2 Reshaping when traffic may contend

Garrido Abenza et al. study video delivery over Wireless Access in Vehicular Environments (WAVE) multi-channel operation and show that the IEEE 1609.4 Control Channel (CCH) / Service Channel (SCH) alternation can accumulate packets during the CCH interval and release them as a burst at the start of the SCH interval, inflating

Table 4 – Comparative summary of related approaches and this dissertation.

Approach	Key mechanism	Requirements	Target traffic	Advantages (claimed)	Relation to this work
TSNCtl (FERAUDO; GARBUGLI; BELLAVISTA, 2024)	App-level slot schedule + FSM	Sync, platoon master	Platoon V2V	Lower collisions vs CSMA/CA	Sync/leader; preprint
PRP-MAC (LI et al., 2024)	Preemptive-resume priority + arbiter	UAV/RSU-assisted stack	BSM uploads	Lower BSM delay	Infra + richer MAC
CFC-MAC (LINN; LIU; GAO, 2023)	Cooperative TDMA + slot repair	Slot sync, coop. signaling	VANET safety	Low collision, determ. access delay	TDMA contract
NR-V2X Mode 2 (THABET; BENDARY; ELBAYOUMY, 2025)	Adaptive SPS / resource selection	Cellular sidelink	Periodic safety	Better resource use vs legacy SPS	Different radio access technology
SkipCCH (GARRIDO ABENZA et al., 2024)	Avoid CCH-induced SCH bursts	IEEE 1609.4 multi-channel	Video + QoS ACs	Less contention, less low-AC starvation	Multi-channel video
Adaptive EDCF (ROMDHANI; NI; TURLETTI, 2003)	CW/persistence retuning	Continuous measurement	Differentiated ACs	Better differentiation under load	Always-on, aggregate
DCC (ETSI, 2018)	CBR-driven TPC/TRC/access	Always-on ITS-G5 loop	Aggregate load	Congestion management	Complementary
Wireless TSN / OFDMA (SELIEM; ZAHRAN; PESCH, 2023; SCHNEIDER et al., 2024)	TAS/TWT/RU scheduling	Wi-Fi 6/NIC mods or sync	Industrial / scheduled	Tighter latency control	Out of OCB scope
This work	**DSCP→VO + event-triggered BE suppress**	**Local OCB EDCA only**	**Crash VO vs BE multicast**	**Protection–cost frontier**	**No sync/infra**

Source: Prepared by the author from the cited works.

contention and loss (GARRIDO ABENZA et al., 2024; IEEE, 2016). Their SkipCCH method reshapes transmission timing so that the MAC avoids those artificial bursts, improving reconstructed video quality and reducing starvation of lower-priority access categories under QoS operation (GARRIDO ABENZA et al., 2024).

SkipCCH is not a crash-alert protocol: it targets video over multi-channel WAVE, not one-hop multicast warnings on a single channel. Its conceptual contribution to this

dissertation is the demonstration that *when* traffic is allowed to contend can matter as much as *which* access category contends (GARRIDO ABENZA et al., 2024). Event-triggered BE suppression is a different reshaping: instead of shifting packet release across channel intervals, it withholds BE channel-access requests while Voice activity associated with a crash is detected. That philosophy also differs from the always-on congestion loops of adaptive EDCA and DCC discussed in Section 3.3: those mechanisms retune aggregate load continuously, whereas BE suppression is armed only for the crash-alert window and then released.

3.2.3 Protection versus cost as a first-class metric

Surveys of IEEE 802.11 QoS mechanisms emphasize that preferential treatment for urgent traffic improves its service only at the expense of weaker treatment for lower-priority flows, including possible starvation under sustained high-priority load (MALIK et al., 2015; KOSEK-SZOTT et al., 2013). Malik et al. note that strict priority can drive lower-priority throughput toward zero when high-priority arrivals remain heavy, and that differentiation itself fails once the channel is overloaded beyond a load limit (MALIK et al., 2015). Kosek-Szott et al. likewise stress that EDCA parameters yield differentiation, not bounds (KOSEK-SZOTT et al., 2013).

These observations have two consequences for related-work positioning. First, any crash-protection scheme that suppresses BE is not inventing a new externality; it is making an existing prioritization cost explicit, bounded, and measurable. Second, evaluation that reports only Voice delay improvements without BE delay, reach, or drop statistics is incomplete relative to the QoS literature's own warnings (MALIK et al., 2015). The dissertation therefore treats the protection-versus-cost frontier as a primary result, not as a secondary side effect.

3.2.4 Synthesis

Backoff-freezing analysis explains why dense vehicular contention threatens timely broadcast (WU et al., 2019). SkipCCH shows that controlling contention timing can protect both high-priority and lower-priority flows in a different WAVE setting (GARRIDO ABENZA et al., 2024). QoS surveys insist that protection always has a cost (MALIK et al., 2015; KOSEK-SZOTT et al., 2013). Together they motivate a design that (i) acts when contention is actually crash-relevant, (ii) reshapes who may contend rather than only remapping queues, and (iii) reports BE penalties with the same seriousness as Voice gains.

3.3 ALWAYS-ON LOAD REGULATION: ADAPTIVE EDCA AND DCC

A third family of mechanisms accepts random access but continuously adapts how aggressively stations contend or how much traffic they inject. Adaptive EDCA and ITS-G5 DCC are the most relevant instances for this dissertation. Both are valuable congestion-management tools; neither is a substitute for short-lived, per-class crash protection.

3.3.1 Adaptive EDCA parameter retuning

Romdhani, Ni, and Turletti proposed Adaptive EDCF, which adjusts persistence factors based on estimated collision rates to improve service differentiation under load in ad hoc IEEE 802.11 networks (ROMDHANI; NI; TURLETTI, 2003). Survey work places such schemes in a broader taxonomy of DCF/EDCA enhancements that vary contention windows, interframe spaces, frame lengths, or admission-control rules in response to measurements (NI; ROMDHANI; TURLETTI, 2004; MALIK et al., 2015). The common pattern is *continuous*, measurement-driven retuning of the contention process itself, typically aimed at sustaining differentiation as the aggregate offered load changes.

Adaptive EDCA therefore shares with this dissertation the belief that fixed EDCA parameters are insufficient across operating regimes (ROMDHANI; NI; TURLETTI, 2003; BIANCHI, 2000). It differs in *what* is adapted and *when*. Adaptive EDCF-style controllers modify contention parameters for ongoing traffic classes as a permanent control loop. The crash-aware controller leaves the `edca_only` AIFSN and CW set unchanged and instead temporarily withholds—or, under emergency preemption, drops—BE channel-access requests while crash-related Voice activity is detected. The adaptation is event-triggered and class-selective rather than a standing retuning of aggregate contention parameters (ROMDHANI; NI; TURLETTI, 2003).

3.3.2 Decentralized Congestion Control in ITS-G5

European ITS-G5 deployments standardize DCC at the access layer (ETSI, 2018). As summarized in Chapter 2, DCC measures the CBR and applies transmit power control, transmit rate control, and DCC access control so that each station's contribution to channel load remains inside a target region; reactive and adaptive algorithm variants are defined (ETSI, 2018). In all cases the loop acts continuously on a station's *aggregate* offered load across traffic classes.

DCC and event-triggered BE suppression therefore optimize different objective functions. DCC keeps the long-term operating point away from congestion collapse by throttling how much traffic a station injects as CBR rises (ETSI, 2018). Crash-aware suppression reshapes the mix of traffic that is allowed to contend during a short alert

window, discriminating BE from Voice rather than scaling all traffic uniformly. The mechanisms are complementary: a deployment could run DCC permanently and still arm local BE suppression when a crash alert is active. The evaluation in this dissertation disables DCC deliberately, so that the five MAC policies' protection-versus-cost frontier is isolated; no claim of superiority over DCC is made or supported (ETSI, 2018).

3.3.3 Synthesis

Adaptive EDCA and DCC demonstrate mature, always-on answers to congestion and differentiation under load (ROMDHANI; NI; TURLETTI, 2003; ETSI, 2018; MALIK et al., 2015). They do not answer whether a short, crash-triggered intervention that leaves EDCA parameters untouched can further compress Voice tails under heavy contention, nor what BE cost that intervention imposes when armed under light contention. Those are the questions reserved for the five-policy matrix in Chapter 4.

3.4 SHORT CONTRAST: WIRED TSN, WIRELESS TSN, AND OFDMA

For completeness, this section briefly situates the dissertation against wired TSN, wireless TSN over Wi-Fi, and OFDMA-based scheduling. The treatment is intentionally concise: these technologies are important in industrial and next-generation Wi-Fi contexts, but they are not the operating assumptions of the crash-aware OCB study.

3.4.1 Wired Time-Sensitive Networking

IEEE 802.1Qbv defines the Time-Aware Shaper, which opens and closes egress gates according to a Gate Control List so that scheduled traffic obtains protected transmission windows on bridged Ethernet (IEEE, 2015). Craciunas et al. study the synthesis of such schedules for real-time communication and show that feasible gate schedules depend sensitively on flow sets and timing constraints (CRACIUNAS et al., 2016). Debnath et al. quantify how combining Time-Aware Shaping with credit-based shaping and frame preemption alters delay and jitter, underscoring that hybrid shaper configurations require careful calibration along the path (DEBNATH et al., 2024, 2023).

Wired TSN assumes full-duplex switched Ethernet, path-wide configuration, and typically IEEE 802.1AS-class synchronization. None of those ingredients exists in decentralized 802.11p OCB V2V (JIANG; DELGROSSI, 2008; IEEE, 2015). TSN results are therefore cited as contrast, not as transferable mechanisms.

3.4.2 Wireless TSN over Wi-Fi

Seliem, Zahran, and Pesch implement wireless TSN extensions that combine prioritization, frame preemption, and credit-based shaping on commodity Wi-Fi for industrial Internet-of-Things scenarios, reporting feasibility alongside the need for NIC or

driver support (SELIEM; ZAHRAN; PESCH, 2023). Satka et al. experimentally characterize wireless TSN behavior across Wi-Fi, 4G, and 5G links and highlight physical-layer variability that complicates deterministic claims (SATKA et al., 2023). Schneider, Sofia, and Kovatsch propose time-aware scheduling aligned with Target Wake Time (TWT) cycles for industrial IoT, depending on endpoint participation in schedule synthesis and on Wi-Fi features beyond legacy 802.11p (SCHNEIDER; SOFIA; KOVATSCH, 2022).

These efforts show that TSN-inspired determinism over Wi-Fi is an active research front, but they target industrial BSS or Wi-Fi 6 feature sets rather than vehicular OCB crash multicast. This dissertation does not evaluate TAS, TWT, preemption hardware, or credit-based shaping on the air interface.

3.4.3 OFDMA and scheduled trigger frames

IEEE 802.11ax-era OFDMA enables an access point to allocate Resource Units (RUs) to stations, creating a scheduled uplink/downlink structure that can support tighter latency control than pure EDCA. Karthik and Palaniswamy study RU-based OFDMA scheduling that maps priorities into spectral allocation patterns (KARTHIK; PALANISWAMY, 2018). Schneider et al. investigate deterministic channel access using MU-EDCA parameterization and, in later work, scheduled Trigger Frames aimed at worst-case latency bounds for industrial Wi-Fi use (SCHNEIDER et al., 2023, 2024).

OFDMA scheduling presupposes an infrastructure AP and Wi-Fi 6/7 hardware capable of RU allocation and triggering. Those prerequisites are outside the OCB 802.11p envelope of this study (JIANG; DELGROSSI, 2008; IEEE, 2021). No OFDMA migration path is claimed or evaluated.

3.4.4 Synthesis

Wired TSN, wireless TSN, and OFDMA illustrate how far the field goes when synchronization, infrastructure, or next-generation PHY/MAC features are available (IEEE, 2015; SELIEM; ZAHRAN; PESCH, 2023; SCHNEIDER et al., 2024). The crash-aware problem statement asks a different question: what protection-versus-cost frontier is achievable when those features are *not* available and only local EDCA behavior can change.

3.5 SYNTHESIS AND RESEARCH GAP

Across the surveyed literature, three recurring design philosophies appear. The first seeks *stronger coordination*—slots, preemption under arbitration, cooperative TDMA, or cellular resource selection—and accepts synchronization or infrastructure as the price of reduced collisions (FERAUDO; GARBUGLI; BELLAVISTA, 2024; LI et al., 2024; LINN; LIU; GAO, 2023; THABET; BENDARY; ELBAYOUMY, 2025). The second

reshapes *contention timing* or reports the *cost* of prioritization, showing that when and whom one allows to contend determines both urgent-flow delay and ordinary-flow harm (GARRIDO ABENZA et al., 2024; WU et al., 2019; MALIK et al., 2015). The third regulates load *continuously* through adaptive EDCA parameters or DCC, optimizing aggregate congestion rather than short crash windows (ROMDHANI; NI; TURLETTI, 2003; ETSI, 2018).

What remains unresolved for the crash-aware OCB setting is a local mechanism that (i) uses only standard EDCA queues and an explicit DSCP-to-Voice mapping (SZIGETI; HENRY; BAKER, 2018; IEEE, 2005), (ii) arms additional BE suppression only while crash-related Voice activity is present, (iii) requires no clock synchronization, roadside unit, UAV, or cellular sidelink, and (iv) is evaluated jointly on Voice protection and BE cost across density and load regimes. Existing stronger MACs solve related problems under broader assumptions (FERAUDO; GARBUGLI; BELLAVISTA, 2024; LI et al., 2024; LINN; LIU; GAO, 2023; THABET; BENDARY; ELBAYOUMY, 2025). Adaptive EDCA and DCC solve always-on congestion under complementary objectives (ROMDHANI; NI; TURLETTI, 2003; ETSI, 2018). TSN and OFDMA solve determinism when infrastructure and advanced PHY/MAC features are present (IEEE, 2015; SCHNEIDER et al., 2024).

Chapter 4 instantiates the residual gap as a concrete system model: a highway one-hop multicast scenario, a two-class BE/Voice traffic abstraction with a timed crash window, and five MAC policies—`plain`, `edca_only`, and three adaptive profiles (`stable`, `guarded`, `emergency`)—that map the protection-versus-cost frontier without leaving the standard-compliant, infrastructure-free design envelope.

4 SYSTEM MODEL

Chapter 2 established the standards path from application criticality through DSCP marking and IEEE 802.11 EDCA to a shared contention medium, and Chapter 3 showed that stronger vehicular MAC alternatives typically require synchronization, infrastructure, or a different radio access technology. The residual design point occupied by this dissertation is narrower: protect rare, high-impact crash alerts with an explicit DSCP 46-to-Voice mapping plus local, event-triggered BE suppression on IEEE 802.11p OCB operation, without reservations, clock synchronization, or a new MAC protocol (JIANG; DELGROSSI, 2008; IEEE, 2021; SZIGETI; HENRY; BAKER, 2018).

This chapter formalizes that design point as an abstract system model—what the proposed system *is*, independently of any particular simulator, programming language, or software stack. It defines the decentralized highway network, the synthetic two-class traffic abstraction, the five MAC policies and three-mode adaptive controller, the crash-event timeline, and the modeling assumptions. Concrete numeric bindings for timing, load, radio, and access-control knobs are recorded with the software realization in Chapter 5; how well the system performs under experiment is deferred to the subsequent evaluation.

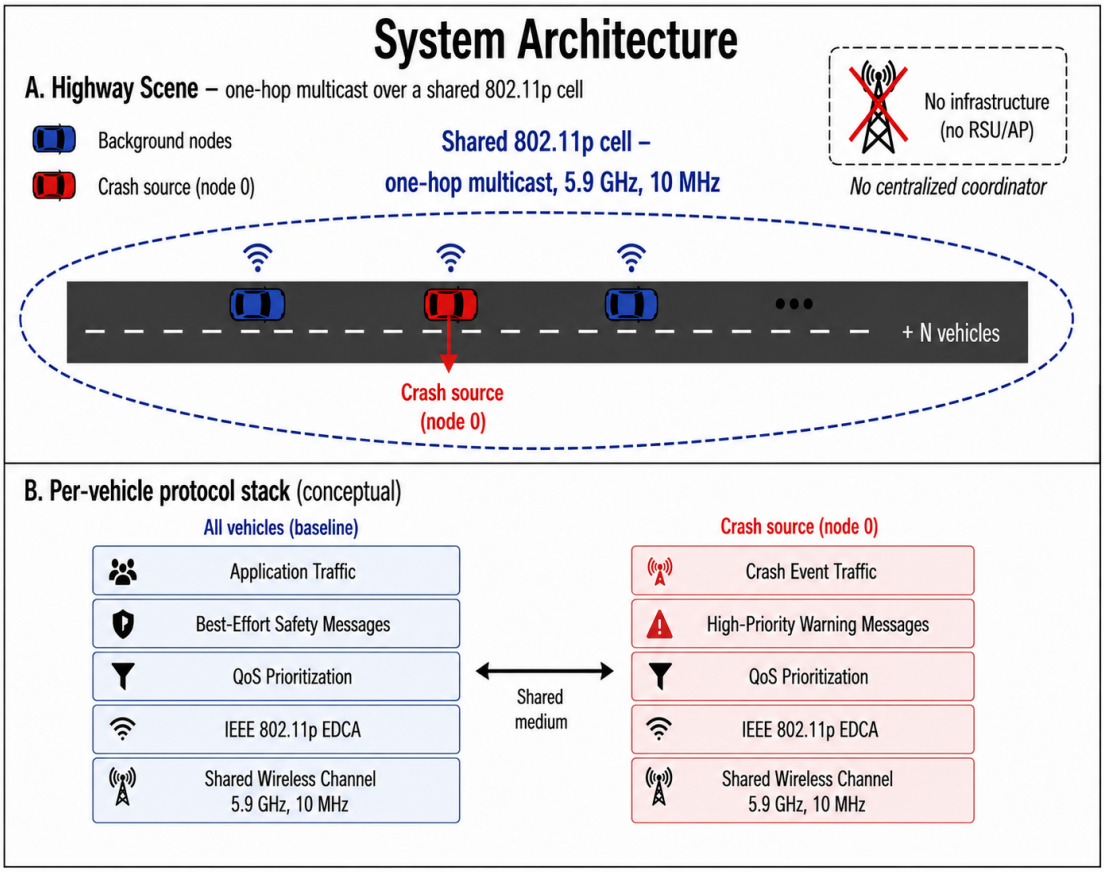
The chapter is organized as follows. Section 4.1 states the network model. Section 4.2 defines the traffic classes. Section 4.3 presents the QoS mapping and the five MAC policies. Section 4.4 describes the crash timeline. Section 4.5 lists assumptions and scope limits.

4.1 NETWORK MODEL

The system considers a decentralized vehicular ad hoc network in which vehicles communicate directly over a shared IEEE 802.11p channel. No access point, roadside unit, or centralized coordinator participates in forwarding or channel allocation (JIANG; DELGROSSI, 2008). All stations therefore contend for the same wireless medium under OCB operation, so the delivery of urgent traffic depends on distributed medium access rather than on reserved transmission opportunities (IEEE, 2021; ETSI, 2020).

At the application level, communication follows a one-hop multicast dissemination model: each transmission is injected into the same contention-based medium and may be received by all vehicles within radio range. Group-addressed IEEE 802 frames inherit the hostile link-layer contract already discussed in Chapter 2—no acknowledgement, no automatic retransmission, and typically a conservative mandatory data rate—so timely delivery hinges on contention control and on correct priority mapping rather than on link-layer recovery (PERKINS et al., 2021). Figure 7 summarizes this infrastructure-free V2V picture.

Figure 7 – System architecture: decentralized vehicle nodes communicating over IEEE 802.11p without infrastructure support.



Source: Prepared by the author.

The radio model uses an IEEE 802.11p physical layer at 5.9 GHz with 10 MHz channelization; reference transmit power, channel index, sensitivity, noise floor, and SNIR threshold are fixed for all policies and recorded with the implementation parameters in Table 12. Propagation is free-space path loss with binary obstruction and without small-scale fading or shadowing, as stated in Section 4.5. *Physical-layer connectivity is held fixed across the five MAC policies*, so differences between BE and VO service reflect queueing and contention rather than time-varying link quality. Dense highway broadcast already stresses this shared medium through repeated backoff freezing under load (WU et al., 2019; BIANCHI, 2000); the model isolates that contention so that the policies can be compared under controlled offered load.

4.2 TRAFFIC MODEL

Traffic uses a synthetic two-class abstraction rather than standardized CAM or DENM schedules, ITS-G5 or WAVE application stacks, or production vehicular logic.

The goal is to isolate MAC-level protection-versus-cost trade-offs under controlled offered load, not to reproduce a complete ITS protocol stack.

Every vehicle generates ordinary background traffic throughout the scenario horizon. This flow represents non-critical cooperative traffic and is modeled by a configurable payload, a random inter-arrival process with a prescribed mean (exponential draws in the software artifact), and default Best Effort marking (DSCP 0). The model instantiates this process with low, medium, and high offered-load profiles so all MAC policies face the same application-side load within a given profile; the concrete means, payloads, and Voice burst parameters appear in Table 12.

Crash-alert dissemination is an event-driven overlay on that background. One designated vehicle becomes the crash source at the configured crash time and begins emitting alert messages marked with DSCP 46, the Expedited Forwarding codepoint used in this study as the crash marker (DAVIE et al., 2002; BABIARZ; CHAN; BAKER, 2006). Each logical alert may be repeated a configurable number of times, with configurable spacing and jitter, to represent short warning bursts; such application-level repetition of safety messages is the reliability mechanism on which robust broadcast schemes for vehicular networks are built (MA et al., 2012; FARNOUD; VALAEE, 2009; YOO; KIM, 2011). Importantly, the crash-alert flow is *superimposed* on the background flow rather than replacing it: the crash source continues to contribute ordinary BE traffic while simultaneously injecting urgent alerts. After the crash window ends, alerts cease and the network returns to BE-only operation, as detailed in Section 4.4.

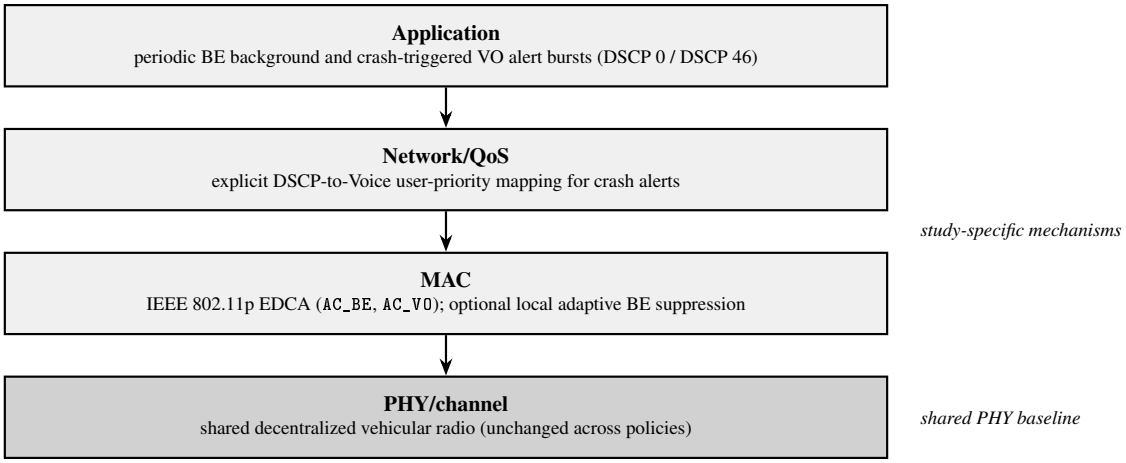
4.3 QOS MODEL AND MAC POLICIES

The model admits only two traffic classes. Periodic background packets remain Best Effort and carry DSCP 0. Crash-alert packets carry DSCP 46 and, in the QoS-enabled policies, are explicitly mapped to the IEEE 802.11 Voice user priority and AC (SZIGETI; HENRY; BAKER, 2018; IEEE, 2005). That mapping is not automatic in every stack; the Background chapter's DiffServ path (Section 2.5) is the normative mapping assumed here for QoS-enabled policies (NICHOLS et al., 1998; BLAKE et al., 1998). Figure 8 summarizes the abstract end-to-end path from application marking to channel access.

Five MAC policies operate on the same traffic and radio model. They form a comparison ladder from undifferentiated access, through standard EDCA differentiation, to three adaptive extensions that temporarily suppress competing BE traffic while crash alerts are active. None of the policies reserves the channel or claims deterministic delay bounds: EDCA provides statistical differentiation only (MANGOLD et al., 2003; KOSEK-SZOTT et al., 2013; KONG et al., 2004).

- `plain` — Non-QoS DCF baseline: undifferentiated access with a single pend-

Figure 8 – Conceptual cross-layer path for crash-aware prioritization. Study-specific mechanisms act at Application, Network/QoS, and MAC; the shared PHY baseline is identical across all policies.



Source: Prepared by the author.

ing queue and no DSCP-to-AC mapping (IEEE, 2021; BIANCHI, 2000).

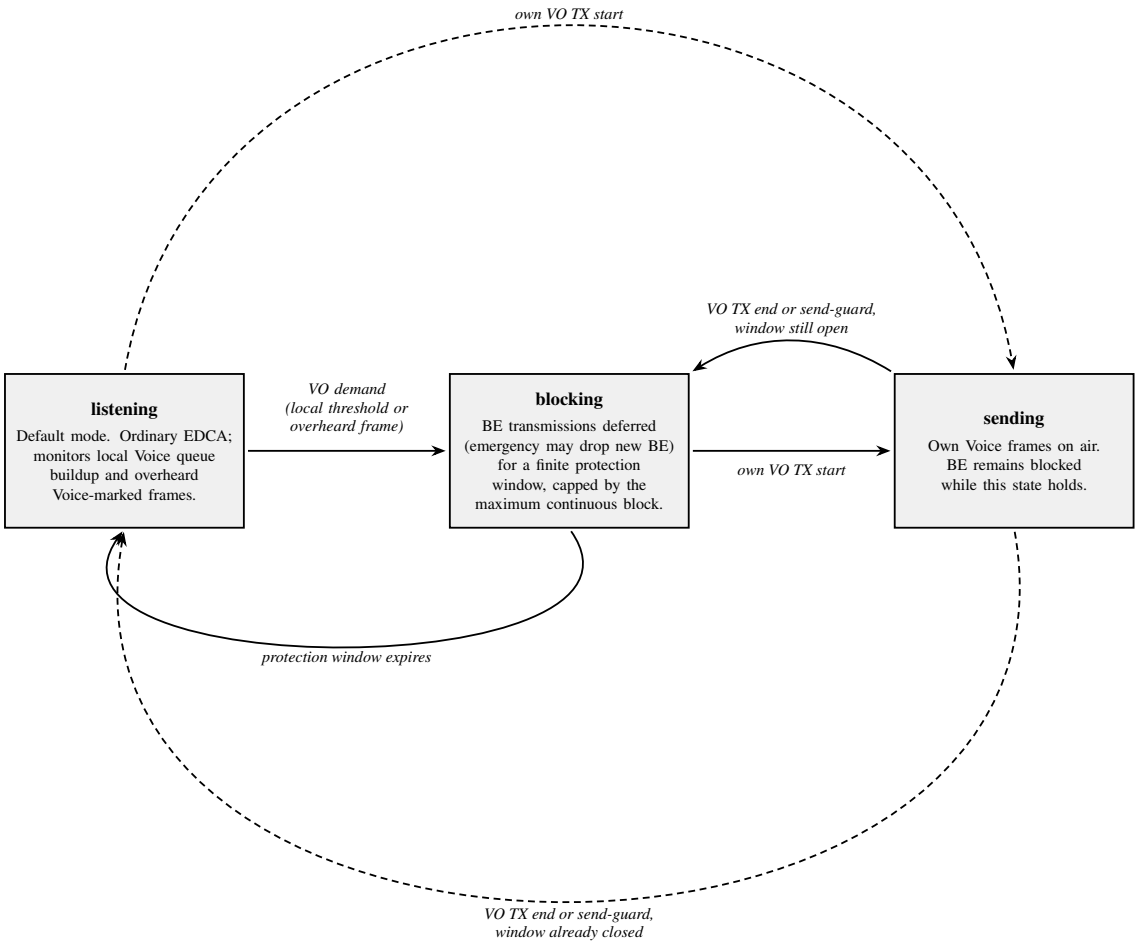
- `edca_only` — Standard QoS baseline: DSCP 46 maps to `AC_VO`; all other packets use user priority 0 (IEEE, 2005).
- `edca_v2x_vo_stable` — Sustained Voice protection: inherits `edca_only` and adds the adaptive controller below with comparatively long protection windows.
- `edca_v2x_vo_guarded` — Bounded suppression: same inheritance with tighter, shorter protection windows.
- `edca_v2x_vo_emergency` — Emergency preemption: same inheritance with aggressive activation; newly arriving BE packets may be dropped while protection is active.

For brevity the adaptive policies are often abbreviated `stable`, `guarded`, and `emergency`. Queue depths, contention windows, and controller timing knobs for each policy are bound in the implementation and listed in Table 12.

4.3.1 Adaptive three-mode controller

The three adaptive policies add a local three-mode Finite-State Machine (FSM)—*listening*, *blocking*, and *sending*—that temporarily gates BE eligibility for channel access without changing the underlying queue parameters inherited from `edca_only`. Figure 9 summarizes the modes and transitions.

Figure 9 – Local adaptive controller modes and transitions for the `edca_v2x_vo_*` policies.



Source: Prepared by the author.

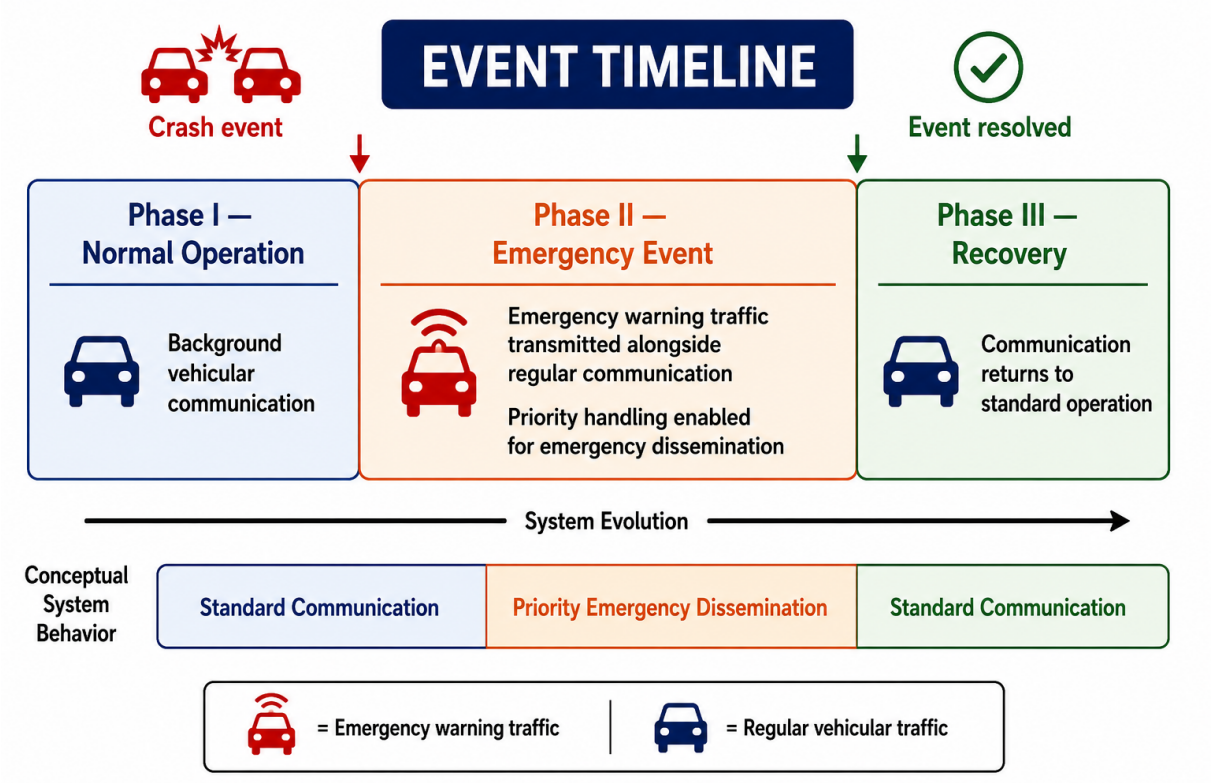
A node leaves *listening* for *blocking* when the local Voice queue crosses the activation threshold or when any Voice-marked frame is overheard—a single overheard alert frame suffices to arm or extend protection; the queue threshold applies only to locally generated Voice traffic. It moves to *sending* when its own Voice frames obtain the channel. After the Voice transmission ends—or if the send-guard timer fires first—it returns to *blocking* while the protection window is still open, and to *listening* once that window has expired. Nodes that only overhear Voice traffic never enter *sending*: they remain in *blocking* until the window expires. Figure 9 also shows a skip from *listening* directly to *sending* when a local Voice frame wins the channel without a

prior demand-triggered block. Under `stable` and `guarded`, BE packets remain queued but lose channel-access eligibility while the controller is blocking or sending. Under `emergency`, newly arriving BE packets may additionally be dropped and stale BE transmission opportunities released. The mechanism shapes local contention during the alert period; it does *not* reserve the medium, open scheduled gates, or provide worst-case latency bounds (IEEE, 2005; KOSEK-SZOTT et al., 2013).

4.4 EVENT MODEL

The system evolves through three phases, illustrated in Figure 10.

Figure 10 – Conceptual event timeline: normal background traffic, crash-triggered Voice prioritization, and recovery.



Source: Prepared by the author.

1. **Normal operation.** Before the crash event, all vehicles generate only periodic BE multicast traffic. No crash-alert Voice flow is present, and adaptive controllers remain in *listening*.
2. **Crash window.** At time 30 s of the scenario horizon, one designated vehicle enters the crash state: it becomes stationary and immediately begins the Voice alert flow for a 30 s window. Ordinary background traffic remains active

across the network, including at the crash source, so urgent packets must compete with existing channel load (PERKINS et al., 2021).

3. **Recovery.** After the crash window expires, crash-alert transmissions cease and the system reverts to BE-only background traffic. Under the adaptive policies, however, the local effect of the crash can briefly outlast the application burst: nodes that still observe Voice queue pressure or receive crash-related Voice frames may remain temporarily in *blocking* or *sending* until the protection window and its short guard interval expire.

This three-phase structure is deliberately simple. It isolates a single crash source and a bounded alert interval so that the five policies can be compared on the same timeline under light and heavy fleet densities.

4.5 ASSUMPTIONS AND SCOPE

The model is intentionally minimal. There is no infrastructure, centralized scheduler, or Time-Sensitive Networking-style gating; there is a single crash source; and there is a two-class BE/VO distinction only. Road geometry, fleet density, and radio parameters fix concrete instances of the model; their reference values are recorded with the implementation in Table 12.

DCC, as standardized for ITS-G5 deployments, is *not* part of the model: application transmit rates follow the fixed offered-load profiles rather than reacting to channel busy ratio (ETSI, 2018). As argued in Chapter 3, DCC remains a complementary always-on regulator; excluding it here isolates the protection-versus-cost behaviour of the five MAC policies.

Small-scale fading, log-normal shadowing, and graded obstacle attenuation are likewise omitted. Connectivity uses free-space path loss, a static SNIR threshold, and binary obstruction (a link is either clear or fully blocked). Contention is therefore induced mainly by multicast load and fleet density, so the policies compete on backoff, queue service, and local BE suppression rather than on cross-layer load regulation or time-varying PHY stress.

Finally, the model makes no deterministic or worst-case latency claims. EDCA and all five policies provide statistical service only (KOSEK-SZOTT et al., 2013; MANGOLD et al., 2003). Quantitative Voice or Best Effort delay statements belong to the evaluation, not to closed-form bounds from this chapter.

Together, these definitions constitute the crash-aware system studied below. Chapter 5 maps them onto the software artifact and its parameter bindings; Chapter 6 then measures Voice service under contention and the Best Effort cost of that protection.

5 IMPLEMENTATION

Chapter 4 defined the crash-aware system at a conceptual level: a decentralized IEEE 802.11p highway network, a synthetic two-class BE/VO traffic model, five MAC policies with a local three-mode controller, and a three-phase crash timeline. This chapter maps that model onto the concrete software artifact developed for this dissertation, so that a reader who already understands the design can reproduce the build, follow the packet path, and locate every study-specific module without consulting undocumented source comments.

The realization is an OMNeT++ 6.1 (VARGA; HORNIG, 2008) / INET 4.5.4 (VARGA et al., 2024) / Veins 5.3.1 (SOMMER; GERMAN; DRESSLER, 2011) project whose source, scenario packages, and analysis tooling are publicly available (AGUIAR; GRACIOLI, 2026). Road mobility is provided by SUMO coupled through TraCI (LOPEZ et al., 2018). The chapter is organized as follows. Section 5.1 pins the toolchain and repository layout. Section 5.2 describes the vehicle host and scenario compound modules. Section 5.3 traces the end-to-end packet path. Section 5.4 details the application modules. Section 5.5 covers the DSCP-to-Voice classifier. Section 5.6 explains the instrumented MAC and the adaptive HCF controller. Section 5.7 documents how the five policies are wired in configuration. Section 5.8 presents the highway scenario packages, the consolidated simulation-parameter table, and the configuration matrix. Section 5.9 lists recorded signals and the analysis tooling surface. Section 5.10 states implementation scope and deliberate omissions. Seeds, statistical aggregation, and measured results belong to Chapter 6 and are not reported here.

5.1 SIMULATION TOOLCHAIN AND REPOSITORY LAYOUT

The Background chapter motivates bidirectional road-and-network coupling for inter-vehicle protocol studies (Section 2.6). Here the same stack is pinned to the versions used for the experiments and narrowed to the modules that realize the System Model.

5.1.1 Framework versions

OMNeT++ 6.1 provides the discrete-event engine, component model, and `.ini`/`.ned` configuration system in which every vehicle node is assembled (VARGA; HORNIG, 2008). INET 4.5.4 supplies the Internet protocol stack, IEEE 802.11p-capable MAC and PHY, radio propagation abstractions, and the HCF into which the study's adaptive controller is substituted for selected policies (VARGA et al., 2024; IEEE, 2005). Veins couples OMNeT++/INET to SUMO through TraCI, so highway mobility, lane discipline, and the crash-node stop command remain consistent with the road

microsimulation (SOMMER; GERMAN; DRESSLER, 2011; LOPEZ et al., 2018). In the artifact tree, `inet/` and `veins/` are git submodules consumed as read-only frameworks, pinned to the INET 4.5.4 and Veins 5.3.1 releases; the study-authored code lives under `veins_qos/` (AGUIAR; GRACIOLI, 2026).

5.1.2 Repository layout

Table 5 summarizes the directories that matter for implementation and reproducibility. Only the active highway packages participate in the dissertation experiment matrix; older square and early highway packages remain in the tree as development history and are not part of the reported study (AGUIAR; GRACIOLI, 2026).

Table 5 – Artifact directory layout relevant to the crash-aware implementation.

Path	Role
`veins_qos/src/traffic/`	Study applications: `CritPacketSender`, `CrashBurstApp`.
`veins_qos/src/qos/`	Explicit DSCP-to-user-priority classifier.
`veins_qos/src/mac/`	Instrumented IEEE 802.11 MAC, adaptive HCF, and FSM controller.
`veins_qos/src/veins_inet/`	Veins↔INET host, mobility, manager, and UDP application base.
`veins_qos/simulations/veins_inet_highway_light/`, `veins_inet_highway_heavy/`	Active density packages: SUMO network, obstacles, `omnetpp.ini`, runners.
`kpi_dashboard/`	Offline parser of OMNeT++ `.sca`/`.vec` result files and figure export.
`inet/`, `veins/`	Framework submodules (read-only).

Source: Prepared by the author.

5.1.3 Attribution

The Veins/INET integration modules under `veins_qos/src/veins_inet/` derive from the Veins project (Sommer et al.) (SOMMER; GERMAN; DRESSLER, 2011). The `VeinsInetTransparentMobility` component incorporates work by Jannusch Bigge (TU Dresden); it is present in the tree but is *not* selected by the active highway configurations, which use `VeinsInetMobility` instead (AGUIAR; GRACIOLI, 2026). Study-specific logic is confined to the application generators, the classifier, the instrumented MAC, and—for the adaptive policies only—a replacement HCF submodule, so that physical-layer connectivity, mobility, and offered-load profiles stay shared across all five MAC policies (Section 4.1).

5.2 HOST AND SCENARIO COMPOUND MODULES

5.2.1 Vehicle host

Each simulated vehicle is an instance of `veins_qos.veins_inet.VeinsInetCar`, a thin extension of INET's `AdhocHost` that carries a single IEEE 802.11 wireless interface (`wlan0`), an IPv4 stack with `HostAutoConfigurator`, and a Veins mobility submodule (VARGA et al., 2024; AGUIAR; GRACIOLI, 2026). The manager launches cars with

```
*.manager.moduleType = "veins_qos.veins_inet.VeinsInetCar"
```

so that every TraCI-injected node receives the same compound type. Mobility is bound to `veins_qos.veins_inet.VeinsInetMobility`, which mirrors SUMO positions, headings, and presence into the INET mobility model (SOMMER; GERMAN; DRESSLER, 2011). IPv4 autoconfiguration joins the multicast group `224.0.0.1` on `wlan0`, matching the application destination used by both traffic generators.

5.2.2 Application base

Both study applications extend `VeinsInetApplicationBase`, itself derived from INET's `ApplicationBase` with Veins-aware User Datagram Protocol (UDP) socket setup (VARGA et al., 2024; AGUIAR; GRACIOLI, 2026). The base class opens a UDP socket on `wlan0`, disables multicast loopback (`setMulticastLoop(false)`), and provides helpers for timed transmission and TraCI access. Study applications therefore inherit a common multicast send path and concentrate their own logic on marking, timing, and instrumentation.

5.2.3 Scenario network

Each active package defines a `Scenario` network Network Description (NED) that composes:

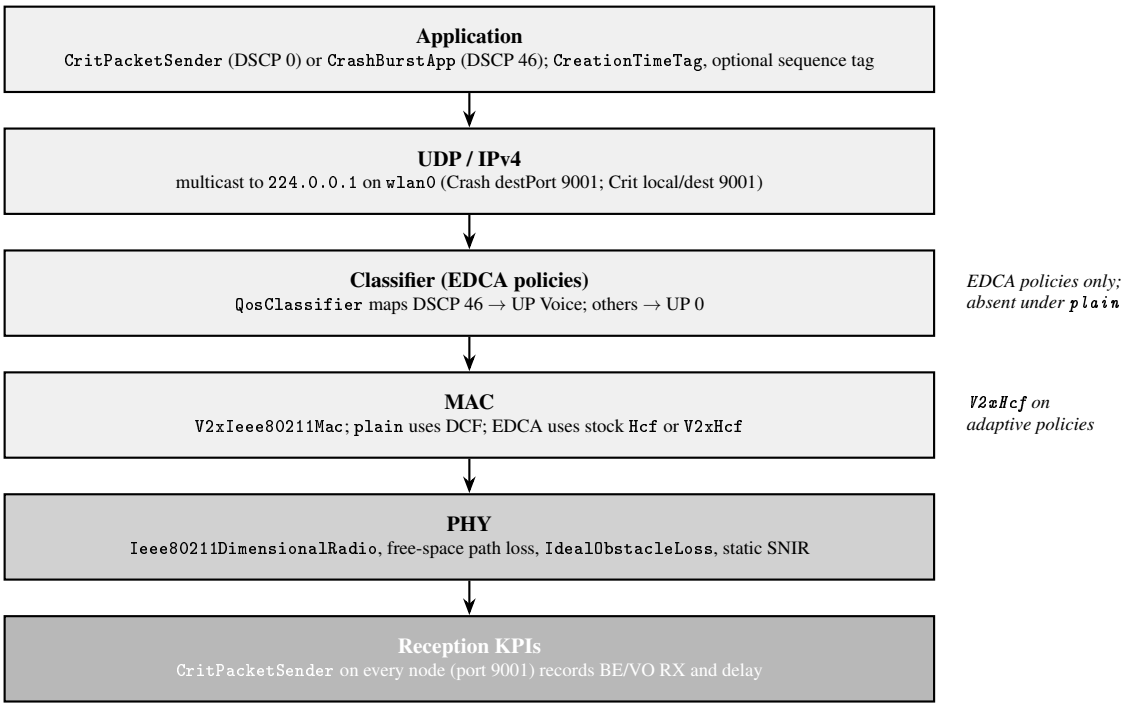
- `Ieee80211DimensionalRadioMedium` — shared radio medium for all nodes;
- `VeinsInetManager` — TraCI scenario manager that launches SUMO through `veins_launchd` on `localhost:9999` using `highway.launchd.xml`;
- `PhysicalEnvironment` — obstacle geometry from `highway.obstacles.xml`;
- visualizers (optional for interactive runs);
- an initial `VeinsInetCar` placeholder; additional cars are injected by TraCI as SUMO vehicles appear.

This compound is the OMNeT++ `network` selected by every runnable configuration in the package (AGUIAR; GRACIOLI, 2026).

5.3 END-TO-END PACKET PATH

Figure 11 and the steps below summarize how a packet moves from creation to KPI recording. The path is identical for Best Effort and Voice except for DSCP marking, classifier mapping, and—under adaptive policies—HCF-level Best Effort gating.

Figure 11 – Implemented cross-layer path from study applications to the shared radio. Adaptive HCF modules appear only under the `edca_v2x_vo_*` policies.



Source: Prepared by the author.

1. The sending application creates a UDP payload, attaches a `CreationTimeTag` (and, for crash bursts, a sequence request), and marks the packet with a `DscpReq` of 0 or 46.
2. `VeinsInetApplicationBase::sendPacket` delivers the frame to UDP, which sends to multicast `224.0.0.1`.
3. IPv4 forwards onto `wlan0`. Under EDCA policies, `QosClassifier` converts DSCP into a `UserPriorityReq` before the frame enters the IEEE 802.11 MAC (SZIGETI; HENRY; BAKER, 2018).
4. `V2xIeee80211Mac` either enqueues into a single DCF pending queue (`plain`) or into EDCA access-category queues (`edca_*`). Adaptive policies consult `V2xEdcaFsmController` before granting Best Effort channel access (IEEE, 2005).

5. The dimensional 802.11p radio transmits on the shared medium; receivers reverse the path through PHY, MAC, IP, and UDP.
6. Every node's `CritPacketSender` (`app[0]`, port 9001) processes received datagrams, classifies them by DSCP into BE or VO KPI streams, skips self-originated addresses, and optionally deduplicates Voice repeats.

Crash-alert frames are therefore generated by `CrashBurstApp` but counted at receivers by `CritPacketSender`, because the crash application sends to destination port 9001—the same port on which every vehicle listens for cooperative traffic (AGUIAR; GRACIOLI, 2026).

5.4 APPLICATION LAYER

Two custom application modules implement the synthetic traffic model of Section 4.2. Both are instantiated on every vehicle (`numApps = 2`); only the configured crash target activates the Voice overlay.

5.4.1 `CritPacketSender`: periodic Best Effort

`veins_qos.traffic.CritPacketSender` extends `VeinsInetApplicationBase` and generates the ordinary background flow (AGUIAR; GRACIOLI, 2026). Table 6 lists the principal NED parameters and recorded statistics.

Table 6 – Principal parameters and statistics of `CritPacketSender`.

Item	Role in the artifact
`sendInterval` `payloadBytes`	Periodic offered load; overridden by `_netload_*` overlays.
`dscp` (default 0)	Best Effort marking carried as `DscpReq`.
`localPort`/`destPort` (9001)	UDP endpoints for BE TX and for all BE/VO RX KPIs.
`voDedupWindow`	Enables run-long Voice deduplication keyed by (source, sequence); the time-valued name is retained for legacy configs, but entries are not mid-run expired.
Statistics	`beTxPackets`, `beRxPackets`, `voRxPackets`, `beEndToEndDelay`, `voEndToEndDelay` (vector + count/mean/min/max).

Source: Prepared by the author.

On transmission, the module stamps `CreationTimeTag` and emits a Best Effort counter. On reception it ignores datagrams whose Layer-3 source equals the local `wlan0` address, maps remaining packets by DSCP into BE or VO delay/RX streams, and—when deduplication is enabled—accepts only the first Voice packet of each (source, se-

quence) pair for the remainder of the run. This design keeps a single receiver module as the KPI sink for both traffic classes.

5.4.2 `CrashBurstApp`: crash-triggered Voice

`veins_qos.traffic.CrashBurstApp` implements the crash-alert overlay (DAVIE et al., 2002; AGUIAR; GRACIOLI, 2026). Every vehicle instantiates the module, but only the node whose OMNeT++ index equals `targetNodeIndex` (default 0) schedules crash behaviour; the sentinel value –1 would activate all nodes and is unused in the active matrix.

At absolute simulation time `crashAt` (30 s in the highway packages), the target node:

1. marks itself visually (red icon) for interactive inspection;
2. commands SUMO through TraCI to set speed to zero, realizing the stationary crash source of Section 4.4 (SOMMER; GERMAN; DRESSLER, 2011; LOPEZ et al., 2018);
3. begins emitting Voice bursts marked with DSCP 46, each logical sequence comprising `repeatCount` physical packets spaced by `repeatGap` with optional `repeatJitter`.

Copy i of a sequence is scheduled at $\max(0, i \cdot \texttt{repeatGap} + U(-J, J))$ after the logical burst time, where J is `repeatJitter`. The jitter therefore applies to the *first* copy as well as to later repeats, so even an alert delivered on its first transmission carries a dispatch offset whose 95th percentile is $0.9\,J$. Because the same logical burst time is the origin of the alert-age metric (Section 5.9.1), this offset is a lower bound on measured Voice delay that no MAC policy can remove; Section 6.5 quantifies its effect on the reported Voice tails.

After `resumeAfter` (30 s), Voice transmission stops and TraCI restores normal speed control (`setSpeed(-1)`). The crash source continues to run `CritPacketSender` throughout, so alerts are superimposed on ongoing Best Effort traffic rather than replacing it (Section 4.2).

Table 7 summarizes the crash-application surface. Physical Voice transmissions increment `voTxPackets`; each logical burst sequence increments `voLogicalTxPackets`. The evaluation prefers logical TX when present so that repeatCount does not inflate the denominator of reception ratios; the implementation exposes both counters precisely for that reason (AGUIAR; GRACIOLI, 2026).

There is no multi-level criticality ladder and no separate four-priority application family: the implementation stays on the two-class BE/VO abstraction fixed in the System Model.

Table 7 – Principal parameters and statistics of `CrashBurstApp`.

Item	**Role in the artifact**
`targetNodeIndex`	Which OMNeT++ node becomes the crash source (default 0).
`crashAt` `resumeAfter`	Absolute crash start and crash-window duration.
`sendInterval` `payloadBytes`	Inter-burst period and payload during the window.

`repeatCount`
`repeatGap`
`repeatJitter`
	Physical repeats of each logical alert. Jitter is drawn per copy, including the first, so every copy carries a dispatch offset relative to the logical burst time.

`localPort` 9002 `destPort` 9001	TX on a dedicated local port; RX KPIs collected by Crit on 9001.
Statistics	`voTxPackets` (physical), `voLogicalTxPackets` (per sequence).

Source: Prepared by the author.

5.5 QOS CLASSIFIER

Correct placement of crash alerts into the Voice access category is not automatic under the common three-MSB DSCP-to-User Priority (UP) heuristic (Section 2.5.3, Section 4.3). The artifact therefore installs an explicit classifier rather than relying on default DiffServ-to-802.11 mapping (SZIGETI; HENRY; BAKER, 2018).

`veins_qos.qos.QosClassifier` implements INET's `IIeee8021dQosClassifier` interface. It reads DSCP from packet indications or requests (and, if needed, from IPv4/IPv6 headers), maps `crashDscp` (default 46) to Voice user priority, and assigns `defaultUp` (default 0, Best Effort) to every other packet (SZIGETI; HENRY; BAKER, 2018; IEEE, 2005; AGUIAR; GRACIOLI, 2026). The resulting `UserPriorityReq` selects the corresponding EDCA access category inside the HCF.

The classifier is installed only under EDCA policies:

```
*.node[*].wlan[*].classifier.typename = "veins_qos.qos.QosClassifier"
*.node[*].wlan[*].classifier.crashDscp = 46
*.node[*].wlan[*].classifier.defaultUp = 0
```

Under `plain`, no classifier is configured: a single DCF pending queue carries all traffic without DSCP-to-AC differentiation, matching the non-QoS baseline of Section 4.3.

5.6 MAC LAYER: INSTRUMENTED MAC AND ADAPTIVE HCF

All active configurations replace the stock IEEE 802.11 MAC typename with an instrumented subclass, while only the adaptive policies replace the HCF submod-

ule (IEEE, 2005; VARGA et al., 2024; AGUIAR; GRACIOLI, 2026).

5.6.1 `V2xIeee80211Mac`

`veins_qos.mac.V2xIeee80211Mac` extends INET's `Ieee80211Mac` and is selected globally:

```
*.node[*].wlan[*].mac.typename = "veins_qos.mac.V2xIeee80211Mac"
```

The module subscribes to `packetDroppedSignal` and, at finish time, records scalar drop totals partitioned by access category (BK, BE, VI, VO, and unclassified) and by drop reason (queue overflow, retry-limit reached, congestion, incorrectly received, and related INET reasons). Those counters are the MAC-side counterpart to the application delay and reception statistics: they expose where frames were discarded under each policy without requiring Evaluation-side reinterpretation of raw event logs (AGUIAR; GRACIOLI, 2026).

5.6.2 `V2xHcf` and `V2xEdcaFsmController`

The adaptive policies substitute

```
*.node[*].wlan[*].mac.hcf.typename = "veins_qos.mac.V2xHcf"
```

for the stock INET `Hcf`. `V2xHcf` implements the `IHcf` contract and embeds a full QoS HCF submodule tree (EDCA, HCCA placeholder, originator/recipient QoS data services, rate selection, acknowledgement and RTS/CTS policies) plus a study-specific `FSMController` of type `V2xEdcaFsmController` (IEEE, 2005; AGUIAR; GRACIOLI, 2026). Because only the typename changes, queue depths and contention parameters inherited from `edca_only` remain numerically identical across the adaptive profiles.

Table 8 lists the controller-facing parameters declared on `V2xHcf` and its FSM submodule. Their semantics match the three-mode controller of Section 4.3.1; this section records how those knobs are named in software.

5.6.3 Controller integration hooks

Table 9 summarizes how `V2xHcf` couples application- and peer-observed Voice activity to Best Effort eligibility. The controller does not rewrite backoff arithmetic or introduce scheduled gates; it interposes on channel-access eligibility for Best Effort while Voice frames remain subject to ordinary EDCA contention (IEEE, 2005).

Table 8 – Adaptive controller parameters exposed by `V2xHcf` / `V2xEdcaFsmController`.

Parameter	**Meaning**
`adaptiveBlocking`	Enables Voice-driven Best Effort suppression through the FSM.
`emergencyPreemption`	When true, newly arriving Best Effort may be dropped and stale Best Effort grants suppressed while protection is active.
`blockDuration`	Protection-window extension per Voice demand event (local enqueue or overheard Voice frame).
`maxContinuousBlock`	Cap on one continuous blocking/sending interval (also mirrored on the FSM).
`voQueueThreshold`	Local Voice queue depth that can trigger protection from upper traffic.
`FSMController.` `sendingGuardTimeout`	Fallback timer armed on Voice TX start; if TX end does not fire first, leave *sending* according to whether the protection window is still open.

Source: Prepared by the author.

Table 9 – Principal integration hooks of `V2xHcf` relative to the stock INET HCF.

Hook	**Trigger**	**Effect**
Voice demand detection	Local Voice enqueue above `voQueueThreshold`, or an overheard Voice-classified MAC frame.	Calls `onVoDemandDetected`; may start or extend a protection window of length `blockDuration`, capped by `maxContinuousBlock`.
Upper-frame gating	FSM reports Best Effort blocked (*blocking* or *sending*).	Best Effort channel-access requests are deferred.
Own Voice on air	Start and end of the node's own Voice transmissions.	On TX start, FSM enters *sending*. On TX end (or send-guard timeout), it returns to *blocking* if the window is still open, otherwise to *listening*.
Emergency preemption	`emergencyPreemption` is true and protection is active.	Newly arriving Best Effort frames may be dropped with a congestion reason; stale Best Effort grants suppressed.
Instrumentation	Controller and FSM events during a run.	Signals/scalars `beDroppedWhileBlocked`, `beGrantSuppressedWhileBlocked`, `voProtectionActivation`; FSM vector `v2xState` (0 = listening, 1 = blocking, 2 = sending).

Source: Prepared by the author.

5.7 MAC POLICY CONFIGURATION WIRING

Table 10 restates the stack mapping. The subsections below document the exact OMNeT++ configuration composition used in the active highway `omnetpp.ini` files (AGUIAR; GRACIOLI, 2026). Numeric access-control values are collected in Table 12; here the emphasis is on *which* typename and parameter paths bind them.

Table 10 – Artifact mapping per stack layer.

Layer	Reused baseline	Introduced modules
App.	Multicast UDP hooks in `VeinsInetApplicationBase`	`CritPacketSender`, `CrashBurstApp`; KPI tags and deduplication
Net/QoS	IPv4 forwarding and transport	`QosClassifier` (DSCP 46 → Voice UP)
MAC	IEEE 802.11p DCF/EDCA queues and channel access	`V2xIeee80211Mac`; `V2xHcf+V2xEdcaFsmController` on adaptive policies
PHY	Dimensional radio, propagation, mobility, TraCI coupling	None (shared across all policies)

Source: Prepared by the author.

5.7.1 `plain`

`[Config plain]` keeps a non-QoS station. It sets the single DCF pending-queue capacity to 128 packets and does not enable `qosStation`, does not install `QosClassifier`, and does not override `hcf.typename`. DSCP markings still travel with IP packets for KPI classification at receivers, but they do not drive EDCA access categories.

5.7.2 `edca_only`

`[Config edca_only]` extends the plain baseline with standard EDCA (IEEE, 2005):

- `mac.qosStation = true`;
- `mac.hcf.typename = "Hcf"` (stock INET);
- `QosClassifier` as above;
- pending-queue capacities BK/BE/VI/VO = 128/128/64/32 (`edcaf[0..3]`);
- MSDU aggregation policy cleared (`msduAggregationPolicy.typename = ""`);
- AIFSN BK/BE/VI/VO = 7/3/2/2;
- `cwMin`/`cwMax` BK/BE = 15–1023, VI = 7–15, VO = 3–7.

INET indexes `edcaf[0]=BK`, `[1]=BE`, `[2]=VI`, `[3]=VO`; the active applications exercise only BE and VO, so VI/BK queues exist but remain empty under the two-class traffic model. Clearing the aggregation policy disables MAC Service Data Unit (MSDU) aggregation, so each application packet maps to exactly one transmitted frame and MAC-level counters stay comparable across policies.

5.7.3 Adaptive profiles

Each adaptive configuration extends `edca_only`, replaces `hcf.typename` with `veins_qos.mac.V2xHcf`, and sets `adaptiveBlocking = true`. Table 11 records the profile-specific knobs exactly as bound in `omnetpp.ini`.

Table 11 – Adaptive policy knobs as bound in the highway `omnetpp.ini`.

Config	block Duration	maxContinuous Block	sendingGuard Timeout	voQueue Threshold	emergency Preemption
`stable`	15 ms	80 ms	5 ms	2	false
`guarded`	4 ms	20 ms	4 ms	3	false
`emergency`	10 ms	60 ms	5 ms	1	true

Source: Prepared by the author.

Legacy configuration names from earlier prototypes (for example Friendly/Protect aliases documented in the artifact's deprecation notes) are not part of the active matrix and must not be confused with the five policies of Section 4.3.

5.8 SCENARIO PACKAGES, SIMULATION PARAMETERS, AND CONFIGURATION MATRIX

5.8.1 Consolidated simulation parameters

Table 12 collects the concrete timing, topology, radio, application, classifier, and MAC bindings used by the active highway artifact. It is the numeric counterpart of the System Model in Chapter 4: the conceptual roles of the five policies and the crash timeline are defined there; the values that instantiate them in this study are recorded here (AGUIAR; GRACIOLI, 2026). Best Effort inter-arrivals are Poisson: OMNeT++ `exponential(·)` is marked `volatile` and redrawn independently before every packet, so the mean is a rate rather than a per-vehicle constant period. Voice burst periods are fixed, with optional per-repeat jitter. Adaptive policies are abbreviated `stable`, `guarded`, and `emergency` in the first column; the full OMNeT++ configuration names are `edca_v2x_vo_*`.

Table 12 – Simulation parameters used in the crash-aware Veins/INET artifact.

Common highway setting		
Parameter	**Value**	**Notes**
Sim. time	70 s	`sim-time-limit`; identical for every configuration.
Corridor	5 km, 3+3 lanes	Straight bidirectional SUMO corridor, three lanes per direction; mixed car/truck types (EIDM, LC2013); nominal lane speed 120 km h^{-1}.

Table 12 – Simulation parameters used in the crash-aware Veins/INET artifact. (continued)

Parameter	**Value**	**Notes**
Density	10 / 100 veh.	Light and heavy packages (`veins_inet_highway_light/_heavy`); same `omnetpp.ini`, different `highway.rou.xml`.
Crash	30 s–60 s	Node index 0 (`node0_highway`): TraCI stop, DSCP 46 alerts; BE background continues.
Radio	802.11p / ch. 3	5.9 GHz, 10 MHz, 20 mW; `Ieee80211DimensionalRadio`; mode `p`; data frames at 12 Mbit/s (fastest mandatory mode, INET default rate selection).
Propagation	FS + binary	Free-space path loss; no fading/shadowing; `IdealObstacleLoss` + `highway.obstacles.xml`; SNIR 4 dB; sens. –85 dBm; noise –110 dBm (INET 4.5.4 defaults, not overridden).
Multicast	`224.0.0.1`	On `wlan0`. Crit ports 9001/9001; Crash local 9002, dest 9001 (VO RX counted in Crit).
TraCI	port 9999	`localhost`; `VeinsInetManager` + `veins_launchd`; update 1 s; SUMO step 0.1 s.
Host/MAC	`VeinsInetCar`	Always `V2xIeee80211Mac`; adaptive policies replace `Hcf` with `V2xHcf`.
Classifier	DSCP 46 → VO	`QosClassifier` on EDCA policies only (`crashDscp=46`, `defaultUp=0`); unused under `plain`.
Recording	vec. + sca.	`**.vector-recording/scalar-recording = true`.
Matrix	15 / package	Five policies × three loads; × two densities = 30 named experiments before seeds.
Application load profiles (`_netload_*`)		
Profile	**BE (`app[0]`)**	**VO crash (`app[1]`)**
`low`	exp. 500 ms, 200 B	120 ms, 150 B; repeat 3; gap 20 ms; jitter 5 ms.
`medium`	exp. 250 ms, 320 B	75 ms, 180 B; repeat 4; gap 15 ms; jitter 5 ms.
`high`	exp. 125 ms, 420 B	20 ms, 260 B; repeat 8; gap 5 ms; jitter 2 ms.
MAC policies (runnable name `<policy>_netload_<profile>`)		
Policy	**Role**	**Access-control parameters**
`plain`	Non-QoS DCF	Single pending queue 128; no classifier; no EDCA; no adaptive blocking.

Table 12 – Simulation parameters used in the crash-aware Veins/INET artifact. (continued)

Parameter	Value	Notes
`edca_only`	Stock EDCA	`Hcf` + `QosClassifier`. DSCP 46 → `AC_VO`, else UP 0. Queues BK/BE/VI/VO: 128/128/64/32. AIFSN 7/3/2/2. `cwMin` 15/15/7/3. `cwMax` 1023/1023/15/7.
`stable`	Sustained VO	Config `edca_v2x_vo_stable`. Inherits `edca_only`; `V2xHcf`. Block 15 ms; max cont. 80 ms; guard 5 ms; VO thr. 2.
`guarded`	Bounded VO	Config `edca_v2x_vo_guarded`. Same inheritance. Block 4 ms; max cont. 20 ms; guard 4 ms; VO thr. 3.
`emergency`	Emergency VO	Config `edca_v2x_vo_emergency`; `emergencyPreemption=true`. Block 10 ms; max cont. 60 ms; guard 5 ms; VO thr. 1; may drop BE.

Source: Prepared by the author.

5.8.2 Density packages

The dissertation experiments use two parallel packages under `veins_qos/simulations/` (AGUIAR; GRACIOLI, 2026):

- `veins_inet_highway_light` — light-density corridor with 10 vehicles;
- `veins_inet_highway_heavy` — heavy-density corridor with 100 vehicles.

Both packages share the same corridor geometry (`highway.net.xml`), obstacle file, launchd description, and—byte-for-byte in the active tree—the same `omnetpp.ini`. Fleet size is selected by the SUMO route file (`highway.rou.xml`): a fixed vehicle `node0_highway` departs at time 0 and becomes OMNeT++ node index 0 (the default crash target), while additional vehicles are injected from SUMO flows. Separating density (package) from offered load (ini overlay) is deliberate: the two stress dimensions can be varied independently (AGUIAR; GRACIOLI, 2026).

5.8.3 Road network and mobility

The SUMO network is a straight bidirectional highway corridor of 5 km with three lanes per direction and mixed car/truck vehicle types under EIDM car-following and LC2013 lane-changing models (LOPEZ et al., 2018). Simulation step size is 0.1 s; teleportation on gridlock is disabled (`time-to-teleport=-1`) so that stopped crash nodes remain in place. Veins updates network-node mobility from TraCI; the manager polls on a one-second update interval in the highway ini, while SUMO itself advances at its finer step (SOMMER; GERMAN; DRESSLER, 2011).

5.8.4 Radio and propagation baseline

Every configuration shares the IEEE 802.11p radio profile summarized in Table 12 (IEEE, 2021; VARGA et al., 2024):

- operation mode `p`, band 5.9 GHz, channel 3, bandwidth 10 MHz, transmit power 20 mW;
- radio typename `Ieee80211DimensionalRadio`;
- antenna mobility attached to the vehicle mobility module with a fixed longitudinal/height offset;
- obstacle loss typename `IdealObstacleLoss`, configured from `highway.obstacles.xml` (INET FRAMEWORK, 2024);
- free-space path loss and a static SNIR threshold, without small-scale fading or shadowing (Section 4.5).

Receiver sensitivity (–85 dBm), energy detection (–85 dBm), the SNIR threshold (4 dB), background noise (–110 dBm), free-space path loss, and constant-speed propagation are the INET 4.5.4 defaults for this radio and medium; they are not overridden in the configuration. No rate-control module is configured, so INET's default rate selection transmits data frames—including group-addressed ones—at the fastest mandatory mode of the 10 MHz OFDM mode set (12 Mbit/s). This differs from production deployments, where group-addressed frames are commonly sent at a basic rate (Section 2.5.4); the choice is uniform across all five policies.

Two fidelity caveats apply. First, the artifact realizes 802.11p-like operation through INET's `AdhocHost` (no association or authentication, ad hoc management); it approximates OCB operation rather than literally implementing `dot11OCBActivated` behaviour. Second, the EDCA parameter set of Table 12 follows the IEEE 802.11 BSS QoS defaults, whereas IEEE 802.11 specifies a distinct default EDCA parameter set for OCB operation with a less aggressive Best Effort category (AIFSN 6 instead of 3) (IEEE, 2021; ETSI, 2020). The configured set therefore makes Best Effort compete harder than an OCB-default deployment would, which is conservative for the protection problem studied here.

Because this baseline is identical across policies, cross-policy differences isolate queueing, contention, and local Best Effort suppression.

5.8.5 Timing and default application binding

Under `[General]`, both packages set `sim-time-limit = 70s`, crash start `crashAt = 30s`, and crash duration `resumeAfter = 30s`, matching Section 4.4 and Table 12. Default application parameters in `[General]` match the medium load profile; the `_netload_*` overlays override them for the full matrix.

5.8.6 Load overlays and runnable matrix

Three helper configurations vary only application-side load, using the means and payloads in Table 12. Runnable experiments compose a policy configuration with a load overlay via OMNeT++ `extends`, for example

```
[Config edca_v2x_vo_emergency_netload_high]
extends = edca_v2x_vo_emergency, _netload_high
```

yielding the naming pattern `<policy>_netload_<low|medium|high>`: five policies $\times$ three loads = fifteen named configurations per density package (AGUIAR; GRACIOLI, 2026). Across both light and heavy packages the unique named experiments therefore total thirty before seed repetitions (seed counts belong to Evaluation).

A sixth overlay family, `_hotspot3`, concentrates application load spatially for the hotspot study of Section 6.4: the three vehicles that enter the corridor immediately after the crash source (nodes 1–3) bulk-upload at a 4 ms Poisson mean with 1200 B payloads ($\approx$2.4 Mbit s^{-1} each) for the whole run, independently of the crash window. Composed as `<policy>_hotspot_<low|medium|high>`, with the overlay listed *before* the `_netload_*` section because OMNeT++ keeps the first matching assignment, it adds fifteen more named configurations to the heavy package.

Batch execution helpers in each package (`run`, `run_matrix.sh`) select profiles such as high-load-only cores or the full fifteen-config matrix; they presuppose a running `veins_launchd` on port 9999. Building the study library follows the standard OMNeT++ pattern (`make makefiles && make` under `veins_qos/`) with NED paths that include INET, Veins, and `veins_inet` sources (AGUIAR; GRACIOLI, 2026).

5.9 INSTRUMENTATION AND ANALYSIS SURFACE

5.9.1 Recorded signals

Highway configurations enable comprehensive recording:

```
**.vector-recording = true
**.scalar-recording = true
```

Table 13 collects the study-relevant statistics emitted by the modules described above. These signals are the measurement surface consumed by Evaluation; their statistical treatment (seeds, pooling, confidence reporting) is deferred to that chapter.

End-to-end delay is computed at receivers as simulation time minus `CreationTimeTag`. Best Effort stamps creation at each physical send. Crash Voice stamps creation at the logical burst start (`logicalCreationTime`), so the Voice delay sample is alert age—the time from the first attempt of that sequence until a receiver's

Table 13 – Study-relevant instrumentation exposed by the artifact.

Source	Recorded quantities
`CritPacketSender`	BE TX/RX counts VO RX counts BE and VO end-to-end delay vectors with count/mean/min/max scalars
`CrashBurstApp`	Physical VO TX count Logical VO sequence count
`V2xIeee80211Mac`	Per-AC drop totals Per-reason drop totals (overflow, retry limit, congestion, . . .)
`V2xHcf`	Best Effort drops while blocked Best Effort grant suppressions Voice protection activations
`V2xEdcaFsmController`	`v2xState` vector (listening/blocking/sending)

Source: Prepared by the author.

first accepted copy—rather than the air time of a later physical repeat (AGUIAR; GRACIOLI, 2026). Alert age consequently includes the per-copy dispatch jitter described in Section 5.4.2, which is the intended semantics (the alert is only as timely as the application makes it) but must be kept in mind when absolute Voice delays are read. Receivers also record a crash-node Best Effort slice (`beEndToEndDelayFromCrash`) for packets whose source is the designated crash vehicle, so Voice can be compared with Best Effort from the same sender.

5.9.2 Offline analysis tooling

The companion `kpi_dashboard/` project parses OMNeT++ scalar and vector result files, caches intermediate tables, and exports publication figures (AGUIAR; GRACIOLI, 2026). From an implementation perspective its contract is:

- ingest `.sca`/`.vec` outputs under each scenario's `results/` directory;
- aggregate application delays and counts from `Scenario.node[*].app[0]` (and crash TX from `app[1]`);
- expose controller and drop counters when present under adaptive policies;
- write tabular and figure artifacts consumed by Evaluation.

The dashboard does not alter simulation behaviour; it is a reproducibility aid that turns the instrumentation of Table 13 into analysis-ready products. Interpretation of those products—protection-versus-cost trade-offs, confidence intervals, and discussion—belongs to the Evaluation chapter.

5.10 IMPLEMENTATION SCOPE AND DELIBERATE OMISSIONS

The code base stays aligned with the minimal System Model of Section 4.5:

- a single crash source and a two-class BE/VO distinction only;
- no infrastructure, roadside unit, or Time-Sensitive Networking-style gating;
- no DCC rate control—application rates follow the fixed offered-load overlays;
- no multi-crash orchestration and no standardized CAM/DENM stack;
- no small-scale fading, shadowing, or graded obstacle attenuation beyond binary `IdealObstacleLoss` (INET FRAMEWORK, 2024).

Older packages under `simulations/veins_inet_square/`, `veins_inet_highway/`, and `veins_inet_light/` are retained for development history and must not be mixed with the active light/heavy highway result trees when analysing the dissertation campaigns (AGUIAR; GRACIOLI, 2026).

Together, the toolchain, host compounds, packet path, applications, classifier, instrumented MAC, adaptive HCF, policy wiring, scenario packages, and instrumentation constitute the concrete realization of the crash-aware System Model. Chapter 6 asks whether crash-triggered Voice traffic obtains better wireless service than ordinary Best Effort under contention, and what cost that protection imposes on ordinary traffic.

6 EVALUATION

Chapter 4 defined the crash-aware system conceptually, and Chapter 5 mapped it onto the Veins/INET artifact. This chapter asks whether that realization works as intended: does crash-triggered Voice traffic obtain better wireless service than ordinary Best Effort under contention, and what cost does that protection impose on ordinary traffic?

The evaluation is a protection-versus-cost study, not a claim of deterministic real-time guarantees (KOSEK-SZOTT et al., 2013; IEEE, 2005). Two highway-density packages are used: a light-density sensitivity sweep (10 vehicles) and a heavy-density congestion stress case (100 vehicles), with emphasis on the latter (AGUIAR; GRACIOLI, 2026). Numeric bindings for timing, radio, load, and MAC policies are those of Table 12; this chapter reports outcomes, not module wiring.

The chapter is organized as follows. Section 6.1 states the experimental matrix and seed budgets. Section 6.2 defines metrics and aggregation. Section 6.3 states the headline answer and the two enabling conditions. Section 6.4 presents the primary positive result: Voice recovery under `hotspot_high`. Section 6.5 records the uniform-load null case that makes that result conditional. Section 6.6 prices the Best Effort cost of protection. Section 6.7 discusses interpretation, practical significance, and limitations. Section 6.8 summarizes the conditional takeaway.

6.1 EXPERIMENTAL SETUP

All experiments use the consolidated bindings in Table 12 and the reproducibility scripts in the public artifact (AGUIAR; GRACIOLI, 2026). Both density packages run the full matrix of five MAC policies and three offered-load profiles (15 configurations) with three OMNeT++ repetitions numbered `#0`–`#2`. Each configuration name follows `<policy>_netload_<low|medium|high>` as documented in Chapter 5. Simulation outputs and the scripts that regenerate the tabulated KPIs are released with the public artifact (AGUIAR; GRACIOLI, 2026). A second matrix replays the same fifteen configurations under a hotspot overlay that concentrates Best Effort airtime near the crash source (15 configurations $\times$ 3 seeds = 45 seeded runs; Section 6.4).

Contention is induced by offered load and fleet density rather than by synthetic random-waypoint mobility alone (SOMMER; GERMAN; DRESSLER, 2011). Best Effort inter-arrivals are Poisson: each vehicle redraws an exponential interval with the configured mean before every packet (500 ms/250 ms/125 ms at low/medium/high). At `netload_high` that mean approaches 800 packets/s fleet-wide once all one hundred vehicles have entered; SUMO injects the fleet progressively, so the realized offer in these traces is about 22 thousand Best Effort transmissions per heavy run. The crash source adds overlapping Voice repeats during the alert window (eight physical copies,

5 ms gap at high load), a sustained stream of roughly 400 physical packets/s while the window is active. With a genuine Poisson offer the channel does *not* enter queue overflow: Best Effort P95 stays sub-millisecond under plain DCF and EDCA, and overflow counters remain zero. The policy contrast under that uniform offer is therefore Best Effort cost, not a Voice-tail race under saturation (AGUIAR; GRACIOLI, 2026; WU et al., 2019).

6.2 METRICS AND AGGREGATION

The primary metrics are class-specific end-to-end delay, jitter, multicast reach, and MAC-layer losses; the definitions below apply to the Evaluation tables in this chapter.

End-to-end delay runs from packet creation to reception at each receiving vehicle. Best Effort stamps creation at each physical send. Crash Voice stamps creation at the logical burst start, so Voice delay is *alert age*: time from the first attempt of a sequence until a receiver's first accepted copy, including any wait for a later physical repeat when earlier copies miss. Mean delays aggregate per-run statistics across receivers, while tail and jitter statistics pool all receivers' per-packet samples, excluding locally looped-back copies. The 95th percentile (P95) is the main tail metric; mean absolute jitter summarizes short-term delay variation on the same samples. A crash-node Best Effort slice records delay only for Best Effort packets whose source is the designated crash vehicle, so Voice can be compared with Best Effort from the same sender rather than with the fleet average.

Because every packet is multicast, reception is a fan-out that can exceed one: Voice *RX/alert* divides received alert copies by logical alert sequences; Best Effort *RX/TX* divides received background copies by transmitted packets. MAC analysis reports total drops, drops per application transmission, and per-access-category attribution (Best Effort, Voice, unclassified); the Voice-attributed drop count quantifies crash-frame loss at the MAC layer, summed over all nodes and including receive-side discards of corrupted frames. When the per-category counters sum to twice the aggregate drop counter, the analysis pipeline halves the attributed counts so that categories sum to the aggregate (a double-counting guard in the parser); attribution is unavailable under `plain`, where no classifier assigns access categories. Controller instrumentation adds Voice protection activations, Best Effort drops while blocked, and held Best Effort grants (Chapter 5).

Each repetition yields one run-level KPI row. Configuration summaries in this chapter are arithmetic means over seeds; tail statistics are computed per run over the pooled receiver samples and then averaged across seeds (a mean of per-run percentiles). Headline delay, jitter, and reach tables also report the minimum–maximum range across those seeds. Both densities use three seeds; *no significance tests are*

reported. The numbers are therefore qualitative mechanism checks under a controlled matrix, not proven effect sizes (AGUIAR; GRACIOLI, 2026). Unless noted otherwise, relative improvements are versus `plain`; Voice-attributed MAC-drop comparisons use EDCA because `plain` has no access-category attribution.

6.3 VOICE PROTECTION UNDER LOAD: THE HEADLINE ANSWER

Does crash-triggered Voice traffic obtain better wireless service than ordinary Best Effort under contention? The answer from the full matrix is conditional: emergency preemption recovers Voice service only when (i) near-crash airtime is concentrated in a few senders and (ii) the alert burst cadence is dense enough to keep protection armed for most of the crash window. Table 14 maps the headline cells; Figure 12 plots the same regime contrast.

At heavy `netload_high` every policy sits near 2.2 ms Voice P95—a floor imposed by the 2 ms per-copy dispatch jitter of the alert generator rather than by channel access; the low and medium profiles, whose jitter is 5 ms, sit near 4.6–4.9 ms instead (Section 6.5). Under `hotspot_high` only emergency compresses the inflated tail. Protection is therefore an event-triggered escalation whose gain is set by channel geometry and burst cadence, not an always-on Voice accelerator. Section 6.4 develops the positive cell; Section 6.5 records the uniform null; Section 6.6 prices the Best Effort bill.

Table 14 – Results at a glance (heavy density, three-seed means). Relative Voice gains are versus `plain`. Detail tables follow in the sections named in the last column.

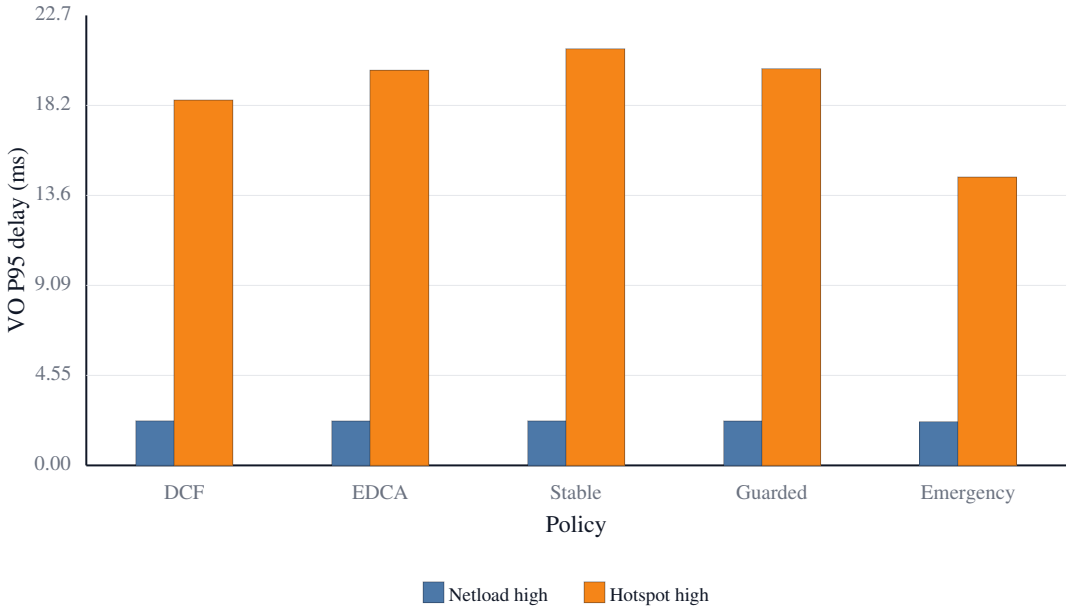
Cell	**Voice P95**	**Main contrast**	**Where**
Uniform `netload_high`	≈2.2 ms, all policies	No Voice win; Best Effort cost	Section 6.5
Hotspot `hotspot_high`	≈18.5 → 14.6 ms	Emergency: –39% mean / –21% P95	Section 6.4
Hotspot `medium`/`low`	No Voice gain	Sparse bursts fail the arming condition	Section 6.4.2
Emergency BE drops	—	$\approx 16.1 \times 10^3$ (hotspot) vs $\approx 2.5 \times 10^3$ (uniform)	Section 6.6

Source: Prepared by the author.

6.4 HOTSPOT LOAD: VOICE RECOVERY UNDER CONCENTRATED CONTENTION

Under a uniform Poisson offer, near-crash collision opportunities are diffuse across about one hundred senders, so node-local Best Effort suppression removes

Figure 12 – Voice P95 delay by policy at high load under uniform Poisson offer versus hotspot contention (100 vehicles, three-seed means). Protection is inert under diffuse load and selective under concentrated near-crash airtime: only emergency compresses the hotspot tail. Here and in the remaining figures the baseline policy is labelled `DCF`; the tables call the same policy `plain`.



Source: Prepared by the author.

a negligible share of ambient airtime and can fire only after a first Voice copy was decoded (Section 6.5). The enabling contraposition is concentrated airtime: if a few crash-adjacent senders dominate the channel, suppression acquires a target and a Voice effect should appear.

The hotspot overlay `_hotspot3` (Section 5.8.6) implements that regime. The three vehicles entering the corridor immediately after the crash node bulk-upload about 2.4 Mbit s^{-1} each for the whole run, roughly tripling aggregate Best Effort transmissions per run from 22×10^3 to 72×10^3. That higher offer pushes Voice-attributed MAC losses from about 3.7×10^3 under uniform `netload_high` to about 25×10^3 under hotspot EDCA *before* policies separate. The crash source, the alert window, and every metric definition are unchanged; configurations are named `<policy>_hotspot_<profile>`.

6.4.1 The `hotspot_high` cell

Table 15 and Figure 13 report the dense-burst high-load cell—the only hotspot cell where Voice delay is policy-sensitive. Plain DCF and plain EDCA reach Voice P95 of 18.5–20 ms. Emergency holds the same tail at 14.6 ms and the Voice mean

at 2.78 ms (–39% mean and –21% P95 versus plain; –38% / –27% versus EDCA). Voice-attributed collision losses fall from 25.2×10^3 to 17.8×10^3 (–30% versus EDCA), and the three-seed P95 ranges do not overlap the baselines'. Emergency lowers both central tendency and tail, but P95 remains an order of magnitude above the mean because alert age is dominated by occasional late first-copy successes under collision-heavy windows. The mechanism is visible in the controller counters: under the 20 ms $\times$ 8-copy alert stream the blocked duty cycle is near-continuous—about 16.1×10^3 Best Effort frames are dropped while blocked (Table 15)—so the hot queues never refill the arena during the window. Multicast reach remains geometry-limited at about ten receivers per alert for every policy: protection buys latency and integrity, not fan-out.

Table 15 – Hotspot high-load Voice protection results (100 vehicles, hotspot nodes 1–3 streaming 1200 B, three-run means). *VO MAC drops* are Voice-attributed incorrectly-received counts summed over nodes; *BE from crash* is the same-sender Best Effort mean delay (starvation under EDCA/`stable`/`guarded`); *BE dropped blocked* counts emergency preemptions.

Policy	VO mean (ms)	VO P95 (ms)	VO RX/alert	VO MAC drops ($\times 10^3$)	BE from crash (ms)	BE dropped blocked ($\times 10^3$)
Plain	4.540	18.45 [16.1–19.9]	9.670	n.a.	1.548	0
EDCA	4.448	19.96 [18.7–20.7]	9.624	25.2	1453 (1.45 s)	0
Stable	4.608	21.04 [18.8–22.8]	9.636	25.4	1386 (1.39 s)	0
Guarded	4.439	20.04 [19.8–20.4]	9.689	25.1	981 (0.98 s)	0
Emergency	**2.781**	**14.56 [11.8–16.0]**	**9.835**	**17.8**	3.335	16.1

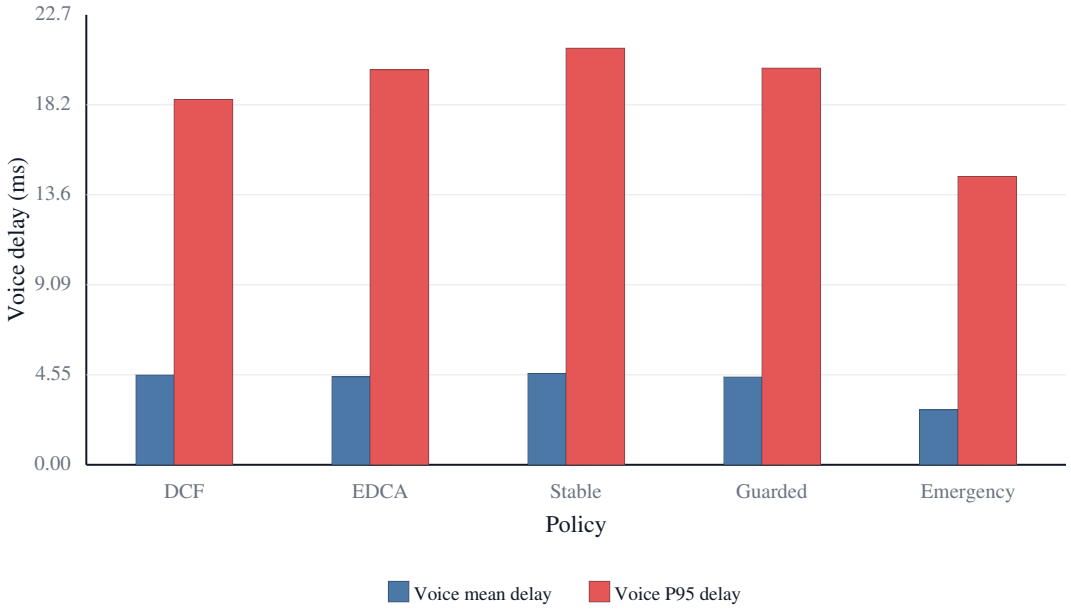
Source: Prepared by the author.

Dropping, not deferring, is the load-shedding step that matters under concentrated contention. `stable` and `guarded` keep Voice P95 on the EDCA baseline (20–21 ms): blocking without dropping never clears the hot queues. The same policies starve the crash node's own Best Effort stream—delivered packets from that sender average 0.98–1.45 s of queueing delay—whereas emergency and plain keep that same-sender slice at 3.3 ms and 1.5 ms respectively (Table 15). Preemption therefore doubles as a starvation cure for the alerted node: it is what both recovers Voice and keeps the sender's ordinary traffic responsive when continuous hotspot demand would otherwise own the local arbitration forever.

6.4.2 Dose response: why only dense bursts win

The Voice gain is conditional on burst density (Table 16, Figure 14). At `hotspot_high` the 20 ms period with eight copies keeps protection armed for most of the crash window. At `medium` (75 ms, four copies) and `low` (120 ms, three copies) the blocked duty cycle falls to roughly one third or less, hot traffic refills the gaps

Figure 13 – Voice mean and P95 delay by policy at `hotspot_high` (100 vehicles). Only emergency, whose preemption keeps hot Best Effort queues out of the collision arena, compresses the tail that hotspot contention inflates.



Source: Prepared by the author.

between blocks, and emergency becomes neutral-to-harmful on the Voice mean (+10% and +16% respectively). Voice service therefore appears only when *both* enabling conditions hold: concentrated near-crash airtime *and* a cadence dense enough to cover it.

Table 16 – Dose response of emergency against plain under hotspot load (three-run means). Voice gains appear only where the burst cadence keeps protection armed for most of the crash window.

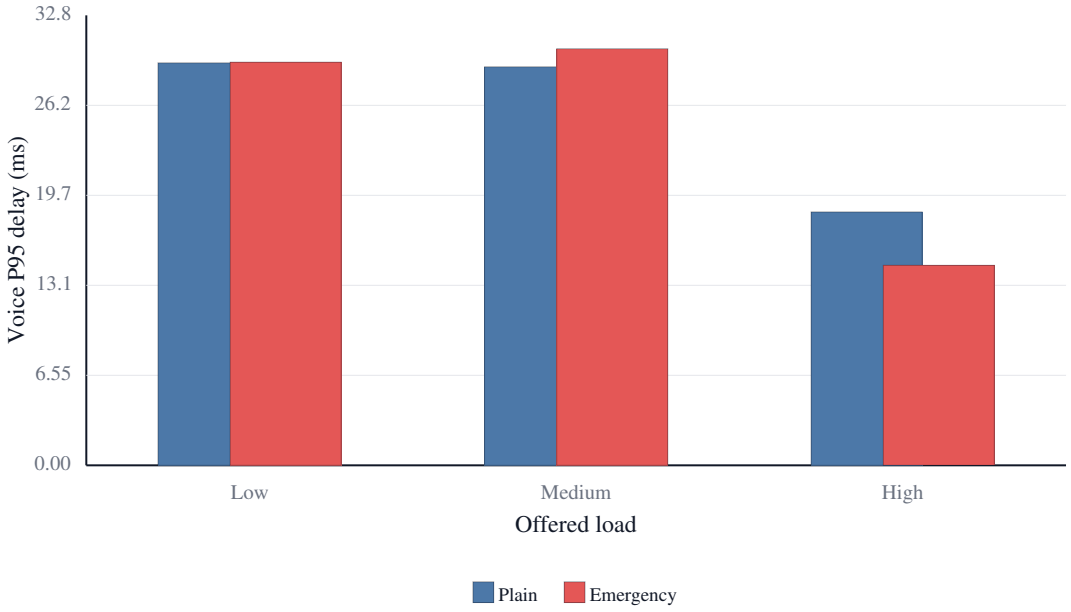
Load	**Burst profile**	**△ VO mean**	**△ VO P95**
`high`	20 ms × 8 copies	–39%	–21%
`medium`	75 ms × 4 copies	+10%	+5%
`low`	120 ms × 3 copies	+16%	+0%

Source: Prepared by the author.

6.5 UNIFORM LOAD: THE NULL VOICE CASE

The uniform Poisson matrix is the scientific null against which Section 6.4 is read. With no concentrated near-crash sender, every policy keeps Voice alert age within a

Figure 14 – Voice P95 delay across offered-load profiles under hotspot load (100 vehicles), plain versus emergency (three-seed means). Protection differentiates only at `high`, where dense bursts keep Best Effort suppression armed for most of the crash window.



Source: Prepared by the author.

few tens of microseconds of plain DCF; the policy contrast is Best Effort cost.

6.5.1 Light-density gating numbers

Table 17 reports light-density high-load cells versus `plain_netload_high`. Voice stays flat near 2.08 ms P95 (within 1.3% of plain). `stable` and `guarded` inflate Best Effort P95 from 0.44 ms to 49 ms and 9.0 ms; emergency shortens it to 0.37 ms by dropping stale Best Effort work rather than by protecting Voice. The sweep is a gating test: adaptive blocking without dropping is already risky when protection windows can open under low channel stress.

6.5.2 Heavy-density confirmation

At one hundred vehicles the same null Voice story holds, only louder. Table 18 and Figure 15 show Voice mean, P95, and jitter within a few tens of microseconds across all five policies (P95 2.21–2.25 ms). EDCA does not lower Voice alert age relative to plain DCF. The policy contrast is Best Effort: `stable` raises fleet Best Effort P95 from 0.51 ms to 30 ms; `guarded` to 3.9 ms; emergency keeps 0.37 ms by preemption. That lower emergency P95 is not better Best Effort service: delay is measured only on delivered packets, so dropping frames while blocked removes the slow tail from the

Table 17 – Light-density high-load gating results versus `plain_netload_high` (10 vehicles, three-seed means). Voice columns are relative changes; Best Effort P95 is absolute (ms) so four-digit percentages do not dominate the table. Plain baselines (ms): Voice mean 0.786, Voice P95 2.078, Best Effort mean 0.383, Best Effort P95 0.437.

Policy	**VO mean**	**VO P95**	**BE P95 (ms)**
Plain (baseline)	—	—	0.44
EDCA	+1.3%	–0.0%	0.42
Stable	+0.9%	+0.5%	49
Guarded	–1.3%	–1.3%	9.0
Emergency	+2.4%	+0.2%	0.37

Source: Prepared by the author.

sample while the cost appears as reach loss (Best Effort RX/TX falls from 8.06 under `plain` to 6.93). Voice RX/alert is 9.95 for every policy (geometry-limited fan-out). Voice-attributed MAC drops sit near 3.7×10^3 under every QoS-enabled policy and do not separate emergency from EDCA; Best Effort queue overflow is zero throughout. Node-local suppression therefore has no Voice target under diffuse contention: it cannot change how many alerts arrive or how many collide.

The floor that every policy shares is set by the alert generator itself rather than by channel access. Each physical copy of a burst—including the first—is dispatched at an offset $\max(0, U(-J, J))$ after the logical burst timestamp that defines the delay origin, where J is the configured repeat jitter (Section 5.4.2). That offset therefore has a point mass at zero and a 95th percentile of $0.9\,J$: about 1.8 ms at the high-load setting $J = 2$ ms and about 4.5 ms at the low and medium setting $J = 5$ ms. Adding the sub-millisecond transfer delay measured on the same samples reproduces the observed floors of about 2.2 ms and 4.6–4.9 ms respectively, and the mean behaves the same way: $\mathbb{E}[\max(0, U(-2,2))] = 0.5$ ms accounts for half of the 0.99 ms Voice mean in Table 18. Only when the first copy is lost does the inter-copy gap g enter, adding one further quantum per missed repeat—the application-level repetition on which robust vehicular broadcast schemes rely (MA et al., 2012; FARNOUD; VALAEE, 2009). At the 95th percentile under diffuse load that regime is rare, so contention shifts probability mass within a generator-imposed floor instead of creating a MAC-policy lever. Two consequences follow for how the 2.2 ms figure should be read: it is not a pure medium-access measurement, and no MAC policy could have moved it.

6.6 THE BEST EFFORT COST OF PROTECTION

Section 6.4 established what emergency preemption *buys* under concentrated load. This section prices what protection *costs*, starting from the hotspot cell and then

Table 18 – Heavy-density high-load delay, jitter, and multicast reach (100 vehicles, three-run means with min–max range across seeds; delays and jitter in ms).

VO (crash alert)				
Policy	**Mean**	**P95**	**Jitter**	**RX/alert**
Plain	0.991 [0.975–1.021]	2.249 [2.229–2.287]	1.024 [1.010–1.044]	9.954 [8.743–10.843]
EDCA	0.994 [0.982–1.003]	2.242 [2.220–2.255]	1.018 [1.005–1.034]	9.954 [8.743–10.843]
Stable	0.988 [0.958–1.026]	2.245 [2.238–2.255]	1.026 [0.995–1.057]	9.954 [8.743–10.843]
Guarded	0.998 [0.991–1.004]	2.241 [2.225–2.253]	1.029 [1.013–1.042]	9.954 [8.743–10.843]
Emergency	0.952 [0.948–0.960]	2.211 [2.203–2.226]	0.979 [0.955–1.007]	9.954 [8.743–10.843]
BE (background)				
Policy	**Mean**	**P95**	**Jitter**	**RX/TX**
Plain	0.387 [0.387–0.388]	0.511 [0.509–0.513]	0.035 [0.034–0.035]	8.057 [7.617–8.321]
EDCA	0.389 [0.388–0.390]	0.524 [0.507–0.538]	0.038 [0.036–0.040]	8.091 [7.687–8.363]
Stable	3.696 [3.560–3.944]	30.491 [27.614–34.175]	3.490 [3.374–3.598]	7.606 [7.295–7.777]
Guarded	0.896 [0.801–0.951]	3.889 [2.955–4.458]	0.669 [0.569–0.720]	7.655 [7.319–7.848]
Emergency	0.432 [0.413–0.461]	**0.369** [0.369–0.369]	0.120 [0.086–0.172]	6.927 [6.692–7.046]

Source: Prepared by the author.

noting the same mechanism under uniform load—where there is no Voice return.

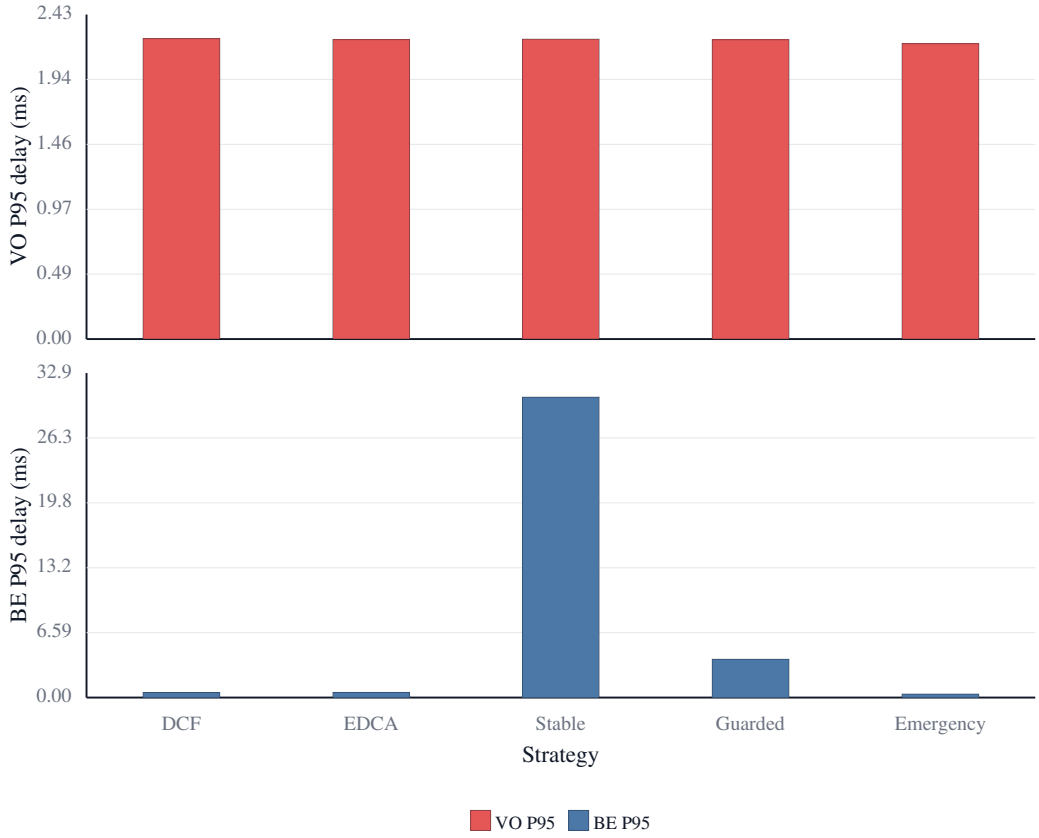
6.6.1 Hotspot pricing

At `hotspot_high` the Best Effort bill is inseparable from the Voice win. Emergency drops about 16.1×10^3 Best Effort frames while blocked (Table 15). That preemption is also what prevents crash-node Best Effort starvation: under EDCA/`stable`/`guarded` the same-sender Best Effort mean reaches 0.98–1.45 s, versus 3.3 ms under emergency. Figure 16 places the tradeoff in Δ-space versus plain DCF: under hotspot high load, emergency alone moves left on the Voice axis; under uniform high load the adaptive policies buy no Voice P95 reduction while Best Effort P95 rises—especially under `stable`.

6.6.2 Uniform-load cost without a Voice target

Under the uniform matrix the same controller still fires, but without a Voice return (Table 19, Table 20, Figure 17). `stable` and `guarded` accumulate about 1.1×10^5 protection activations per high-load run with zero preemption counters, stretching fleet Best

Figure 15 – Voice and Best Effort P95 delay at `netload_high`, heavy density (100 vehicles), dual panel. Top: Voice alert age stays near 2.2 ms for every policy. Bottom: `stable` stretches fleet Best Effort P95 to 30 ms while emergency keeps it at 0.37 ms by dropping competing frames. The horizontal axis, labelled *Strategy* here, carries the same five policies as the *Policy* axis of Figure 12.



Source: Prepared by the author.

Effort P95 to 30 ms and 3.9 ms. Emergency pays about 2.5×10^3 Best Effort drops while blocked plus twelve suppressed grants and keeps fleet and crash-sender tails at 0.37 ms and 0.44 ms—below plain—while Voice P95 remains on the 2.2 ms floor. In short: deferral buys delay; dropping buys silence; only dropping yields Voice service once a concentrated target exists (Section 6.4).

Figure 16 – Voice-gain versus Best Effort cost relative to plain DCF at high load (three-seed means). Dual panels use independent axes: left is uniform `netload_high`; right is `hotspot_high`. Negative Δ Voice P95 is an improvement; positive Δ Best Effort P95 is a penalty.



Source: Prepared by the author.

Table 20 – Adaptive-controller actions at heavy-density high load (three-run means; Voice-protection and Best Effort-drop columns in thousands, $\times 10^3$; Best Effort hold is a raw event count). The `emergency` hold entry is about 12 grant-suppression events.

Policy	VO protection	BE drop while blocked	BE hold
Stable	115	0	0
Guarded	115	0	0
Emergency	127	2.53	12

Source: Prepared by the author.

Table 19 – Best Effort 95th-percentile delay (ms) across offered-load regimes for the five-policy ladder (100 vehicles, three-run means with min–max range across seeds). Voice rows are omitted: Voice P95 stays on the generator's dispatch-jitter floor for every policy (Section 6.5). Cells with a vanishing seed spread omit the range.

Load	Plain	EDCA	Stable	Guarded	Emerg.
BE (background, fleet)					
`low`	0.217	0.217	0.323 [0.217–0.419]	0.217	0.217
`medium`	0.297	0.297	2.081 [0.711–3.203]	0.357 [0.297–0.387]	0.297
`high`	0.511 [0.509–0.513]	0.524 [0.507–0.538]	30.491 [27.614–34.175]	3.889 [2.955–4.458]	**0.369**
BE from the crash node					
`low`	0.217	0.217	0.708 [0.641–0.753]	1.357 [0.217–2.871]	0.217
`medium`	0.297	0.335 [0.297–0.411]	3.040 [2.419–3.814]	2.323 [1.197–3.110]	0.297
`high`	0.631 [0.621–0.651]	0.772 [0.731–0.842]	13.763 [13.541–13.955]	3.865 [3.769–3.944]	**0.435** [0.369–0.567]

Source: Prepared by the author.

6.7 DISCUSSION

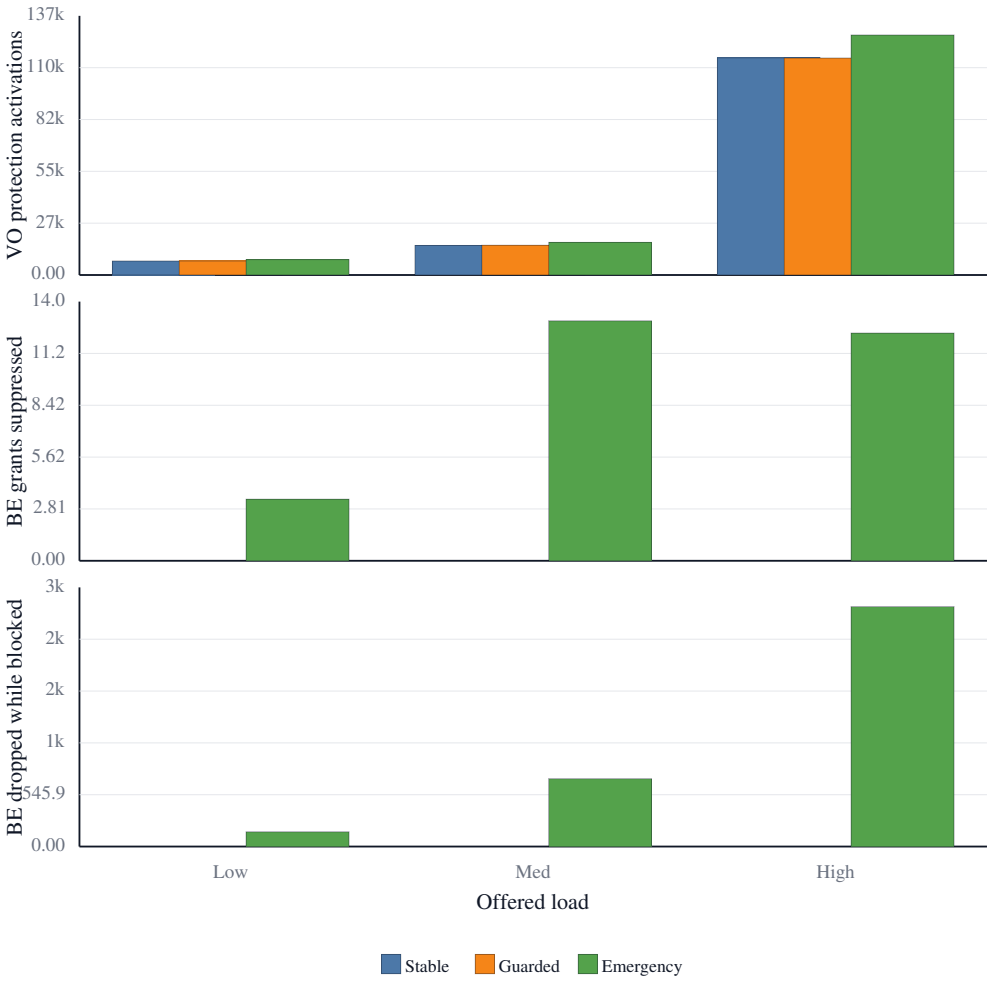
6.7.1 Event-triggered escalation, not always-on QoS

Adaptive blocking is harmful when armed without Voice-limited congestion: `stable` and `guarded` inflate Best Effort delay while Voice alert age stays put (Section 6.5). The mechanism should be treated as an escalation level attached to genuine channel stress, not as an always-on replacement for ordinary EDCA marking (IEEE, 2005; AGUIAR; GRACIOLI, 2026). Explicit DSCP-to-Voice mapping remains a reasonable default; bounded or emergency suppression should be reserved for the concentrated regime of Section 6.4, where preemption flips from pure cost to measured Voice benefit (Table 15).

6.7.2 Reading the baseline correctly

Two measurement traps are aggregation artifacts rather than policy effects. First, under `plain` both classes receive identical per-frame MAC treatment, yet Table 18 shows a Voice–Best Effort delay gap: the gap reflects different traffic processes (one crash-node alert stream versus a fleet-wide Poisson offer), not a differentiating DCF queue. Second, Voice RX/alert exceeds Best Effort RX/TX for structural reasons—different denominators and a denser neighborhood around the stopped crash node. Cross-class comparisons within one policy therefore describe different traffic processes; the crash-node Best Effort slice is the controlled same-sender check, and cross-policy comparisons within one class remain the policy measurement. Poisson redraws keep

Figure 17 – Adaptive-controller actions by offered load, heavy density (three panels): Voice-protection activations, Best Effort grants suppressed while blocked, and Best Effort frames dropped while blocked. Emergency alone combines protection with explicit preemption; `stable` and `guarded` mainly accumulate protection windows without dropping.



Source: Prepared by the author.

Best Effort offer at about 22 thousand transmissions on every uniform run, so seeds no longer split into an uncongested draw and two saturated draws.

6.7.3 Regime-conditional significance of the Voice effects

Under the uniform matrix there is no Voice-tail reduction to claim against representative V2X latency budgets such as the 25 ms end-to-end bound for cooperative platooning in 3GPP TS 22.186 (3GPP, 2020): absolute Voice P95 stays near 2.2 ms for every policy (Section 6.5). That margin is moreover partly structural rather than earned by the channel, since about 1.8 ms of the 2.2 ms is the alert generator's own per-copy dispatch jitter (Section 6.5); the medium-access contribution to the uniform-load tail is

smaller still. The hotspot cell of Table 15 is where absolute tails re-enter that budget band and where emergency compresses them. Three seeds per cell keep this a mechanism check rather than an effect-size estimate (Section 6.7.8); the boundary itself is the take-home—the same preemption is a cost center under uniform load and a Voice lever under concentrated load.

6.7.4 Protection cost, class granularity, and spatial fairness

Bounded protection makes Best Effort service worse than plain DCF at high uniform load, and emergency converts part of the backlog into explicit loss while keeping the fleet Best Effort tail short (Section 6.6). Because the model treats *all* Best Effort traffic as suppressible, these penalties are a worst-case bound for the two-class abstraction (Section 4.5): in a real stack, traffic outside the current crash alert may still be safety relevant. Deployments would need finer-grained classes before comparable MAC aggression, and the trade-off is defensible only for genuinely critical, short-lived flows such as the alert window studied here.

Nor is the penalty uniform across vehicles: protection is armed locally by overheard Voice frames, so vehicles near the crash source defer or drop Best Effort while distant vehicles continue unaffected (Section 4.3.1). The Best Effort cost is spatially concentrated around the event—arguably desirable for a crash—but pooled averages understate the worst-affected vehicles; per-vehicle fairness is left as future work.

6.7.5 Preemption as a starvation cure under concentrated contention

Under concentrated contention, EDCA-family queues can starve the very node they should serve first. At `hotspot_high`, crash-node Best Effort means of 0.98–1.45 s under EDCA/`stable`/`guarded` collapse to 3.3 ms under emergency (Table 15). Dropping is therefore both the Voice lever and the starvation cure when continuous hotspot demand would otherwise own local arbitration.

6.7.6 Burst cadence as an online arming trigger

The dose-response of Table 16 implies a measurable online signal for deciding when protection is worth its cost: arm emergency suppression only while recent alert cadence times the bounded block window (Table 12) saturates the available blocking budget. The controller already tracks activation count and `maxContinuousBlock`, so the trigger is an instrumentation problem, not a new measurement. This closes the escalation design loop of Section 6.7.1.

6.7.7 Denial-of-service risk of Voice-triggered suppression

Protection is armed by local Voice queue pressure and *overheard* Voice frames without authenticating the sender, so a faulty or malicious node continuously emitting Voice-marked traffic could keep nearby vehicles in blocking mode. The risk is bounded but not removed: protection windows are finite (20–80 ms maximum continuous block plus a guard interval, Table 12) and suppression affects only a node's own Best Effort transmissions. Real deployment would additionally require V2X message authentication, rate limiting of alert bursts, and plausibility checks before Voice frames may trigger suppression.

6.7.8 Limitations and threats to validity

Heavy-density KPIs average three seeds and report the min–max range on headline delay tables, still without confidence intervals or significance tests, so results are qualitative mechanism checks rather than proven effect sizes; all observed prioritization is statistical contention shaping, not deterministic real-time communication (KOSEK-SZOTT et al., 2013). The evaluation is simulation-only, models a single crash source on a one-hop highway corridor, and uses the simplified PHY assumptions of Chapter 5 and Section 4.5; concurrent alerts would arm overlapping protection windows whose interaction is not evaluated. Results do not predict absolute performance under richer propagation or congestion control, and no adaptive-EDCA or DCC baseline is implemented, so no superiority claim is made against those families (ETSI, 2018). The configured EDCA parameters follow the IEEE 802.11 BSS QoS defaults rather than the distinct OCB default set, and group-addressed frames use INET's default fastest-mandatory-mode rate selection rather than a basic-rate policy (Section 5.8.4); both choices are uniform across policies but may shift absolute values relative to deployed 802.11p stacks. The `high` crash-burst profile is a deliberate stress case rather than a standardized alert schedule, and no application-level safety deadline is evaluated. The three adaptive profiles are fixed design points whose triggers are not individually ablated, although their spread of block windows (4–15 ms) and Voice thresholds (1–3 packets) acts as a coarse sensitivity ladder; a systematic parameter sweep is future work. The hotspot cells place streaming nodes by injection order rather than by a received-power map, so the exact gain magnitudes depend on that geometry; the sign of the effect—and its absence at sparser burst cadences—is the claim, not the decimals. Within that scope the comparison remains internally fair: all configurations share the same PHY stack, mobility trace, and load matrix, isolating MAC-policy effects (AGUIAR; GRACIOLI, 2026).

6.8 ~~CHAPTER~~ SUMMARY

The evaluation compared five MAC policies on the same highway matrix and found a regime-conditional answer rather than a universal Voice-tail race. The primary positive result is `hotspot_high`: emergency preemption recovers Voice service when near-crash airtime is concentrated and bursts are dense enough to keep protection armed (Table 15). The uniform Poisson matrix is the null that makes that result conditional: Voice alert age stays on the generator's dispatch-jitter floor for every policy, while adaptive blocking mainly taxes Best Effort (Section 6.5). Bounded blocking without dropping (`stable`, `guarded`) never converts hotspot stress into Voice service; dropping is the load-shedding step that both compresses the Voice tail and cures crash-node Best Effort starvation. The practical takeaway is therefore conditional: explicit DSCP-to-Voice marking is a reasonable default; adaptive blocking—especially emergency preemption—is justified as an event-triggered escalation where near-crash congestion is concentrated, with authentication and rate limiting as deployment prerequisites.

The subsequent chapter concludes the dissertation by summarizing these findings, stating the conditional practical takeaway, and outlining limitations and future work.

7 CONCLUSION

This dissertation asked whether crash-triggered Voice traffic can obtain better wireless service than ordinary Best Effort under contention in decentralized vehicular Wi-Fi, and at what cost that protection is paid. The answer was pursued with a deliberately minimal two-class model: all vehicles exchange periodic Best Effort multicast, one designated crash source injects temporary Voice alert bursts marked DSCP 46, and five MAC policies are compared on the same highway matrix—plain DCF, EDCA with explicit Voice mapping, and three adaptive extensions (`stable`, `guarded`, `emergency`) that suppress competing Best Effort while alerts are active (Chapter 4, Chapter 5). The evaluation is simulation-only and mechanism-oriented: it isolates statistical contention shaping under controlled load and density, not deterministic real-time guarantees or application-level crash outcomes (Chapter 6) (KOSEK-SZOTT et al., 2013; IEEE, 2005; AGUIAR; GRACIOLI, 2026).

7.1 MAIN FINDINGS

The primary positive result is the hotspot cell (Table 14, Section 6.4): with three crash-adjacent vehicles bulk-uploading about 2.4 Mbit s^{-1} each and the dense 20 ms burst profile, emergency preemption recovers Voice service — 39% lower mean and 21% lower P95 alert age than plain DCF, with 30% fewer Voice-attributed collision losses — at the price of about sixteen thousand explicitly dropped Best Effort frames per run. The same overlay at sparser alert cadences (`medium`/`low`) restores the null effect, so the enabling conditions are jointly (i) concentrated near-crash airtime and (ii) a burst cadence dense enough to keep protection armed for most of the crash window.

The uniform Poisson matrix is the scientific null against that win. Under heavy density and high offered load, Voice 95th-percentile alert age is about 2.2 ms under every policy: EDCA does not beat plain DCF, and emergency's 2.21 ms versus 2.24 ms is not a material Voice-tail win. The policy contrast there is Best Effort cost: `stable` raises fleet Best Effort P95 from 0.51 ms under plain DCF to 30 ms; `guarded` to 3.9 ms; emergency keeps 0.37 ms by dropping competing frames (about 2.5×10^3 Best Effort drops while blocked) (AGUIAR; GRACIOLI, 2026). The light-density high-load sweep is the same gating test at smaller magnitude: bounded adaptive profiles leave Voice flat while inflating Best Effort P95.

7.2 PRACTICAL TAKEAWAY

The practical takeaway is therefore conditional. Explicit DSCP-to-Voice mapping remains a reasonable default for crash-marked alerts (IEEE, 2005). Adaptive Best Effort suppression—especially emergency preemption—should be treated as an event-

triggered escalation level attached to genuine congestion and genuine alert activity, not as an always-on replacement for ordinary EDCA. Deployment would additionally require authentication, rate limiting, and plausibility checks before overheard Voice frames may arm suppression, because the same local trigger that protects a crash can be abused by faulty or malicious Voice traffic (Chapter 6).

Under a uniform Poisson offer the absolute Voice delays remain far below representative platooning latency budgets such as the 25 ms end-to-end bound in 3GPP TS 22.186 (3GPP, 2020) — and most of the small residual is the alert generator's own dispatch jitter rather than channel access (Section 6.5) — so alert latency is not the risk there; the Best Effort cost is. Under concentrated hotspot contention the same budget becomes meaningful: plain DCF/EDCA reach Voice P95 of 18.5–20 ms, within reach of that bound, while emergency keeps it at 14.6 ms (Section 6.4). The contribution is accordingly regime-aware MAC-level accounting — cost when diffuse, benefit when concentrated — not a claim that the simulated scenario alone guarantees application-level crash outcomes.

7.3 LIMITATIONS

The study's limits are those already argued in the Evaluation discussion (Section 6.7); they are restated here as the boundary conditions on the Conclusion.

Simulation scope and statistical strength. The evaluation is simulation-only, with no testbed or hardware validation, so claims remain tied to the configured Veins/INET model (AGUIAR; GRACIOLI, 2026). Heavy-density KPIs average three seeds and report the min–max range on headline delay tables, still without confidence intervals or significance tests, and light-density results use the same three-seed rule; the numbers are therefore qualitative mechanism checks under a controlled matrix, not proven effect sizes. All observed prioritization is statistical contention shaping, not deterministic real-time communication (KOSEK-SZOTT et al., 2013).

Traffic abstraction and baselines. The traffic model is a synthetic two-class Best Effort / Voice abstraction rather than CAM/DENM schedules or richer multi-class ITS stacks, and the high crash-burst profile is a deliberate stress case rather than a standardized alert timetable. Treating *all* Best Effort as suppressible is a worst-case bound for that abstraction: in a real stack, traffic outside the current crash alert may still be safety relevant, so a binary split can underestimate preemption harm (Section 6.7.4). No adaptive-EDCA or DCC baseline is implemented, so no superiority claim is made against those always-on families (ETSI, 2018; ROMDHANI; NI; TURLETTI, 2003). The three adaptive profiles are fixed design points whose triggers are not individually ablated.

Radio, topology, and delivery model. Connectivity uses free-space path loss, a static

SNIR threshold, and binary obstacle loss; small-scale fading, shadowing, and graded attenuation are omitted (Chapter 5). The corridor is a one-hop highway multicast setting with a single crash source: overlapping protection windows from concurrent alerts are not evaluated, alerts are not retransmitted as multi-hop relays across the 5 km road, and only the light/heavy density pair with three offered-load levels is swept—not a continuous highway-parameter space of speeds, lane layouts, or channel plans.

Interpretation caveats that affect how results are read. The reported 2.2 ms Voice P95 quantifies alert age under a moderate Poisson offer and is dominated by the alert generator's per-copy dispatch jitter rather than by medium access, so it is a floor no MAC policy could have lowered; delays there remain far below representative platooning budgets (3GPP, 2020) and therefore describe mechanism rather than application-level safety benefit; under hotspot contention the baselines approach that budget (18.5–20 ms P95) and emergency keeps 14.6 ms (Section 6.7.3), so absolute claims remain cell-specific, while Best Effort penalties from untimely blocking can still be severe in either cell. Hotspot streaming nodes are placed by injection order rather than by a received-power map (Section 6.7.8), so hotspot gain magnitudes depend on that geometry; placement sweeps belong to future work. Under `plain`, Best Effort and Voice statistics differ because the traffic processes differ, not because the single DCF queue differentiates them (Section 6.7.2); only within-class, cross-policy comparisons are controlled. Pooled Best Effort averages understate spatially concentrated cost near the crash source (Section 6.7.4), and unauthenticated Voice frames can arm suppression—a denial-of-service risk bounded by finite windows but not removed without authentication and rate limiting (Section 6.7.7).

Within that scope the comparison remains internally fair—shared radio stack, mobility, and load matrix—so the reported differences isolate MAC-policy effects (AGUIAR; GRACIOLI, 2026).

7.4 FUTURE WORK

Several extensions follow directly from the limitations above and from the Evaluation discussion.

Stronger evidence and richer scenarios. Larger seed batches with confidence intervals or significance tests would move the study from qualitative mechanism checks toward statistically supported effect sizes. Richer propagation (capture effect, small-scale fading, shadowing, graded obstacles), continuous sweeps of density and offered load, multi-channel or multi-hop relay settings on long corridors, and standardized CAM/DENM-like schedules would test whether the conditional frontier survives more realistic radio and traffic stress. Connecting Voice delay and loss to an application-level safety deadline, crash-avoidance distance, or other outcome metric would address the

practical-significance gap left by the present 2.2 ms Voice tails (3GPP, 2020).

Baselines, parameters, and concurrency. Activation-criteria ablations and systematic parameter sweeps would separate which trigger and window settings drive the observed Voice gains. Adaptive-EDCA, ITS-G5 DCC, and sidelink baselines would place the event-triggered profiles against established always-on congestion and prioritization families (ETSI, 2018; ROMDHANI; NI; TURLETTI, 2003). Concurrent alert sources—and the interaction of overlapping protection windows—remain unevaluated. A closed-loop arming criterion based on the blocked duty cycle (alert cadence times bounded block window, Section 6.7.6) would let the controller engage emergency suppression only under the coverage conditions where Table 16 shows a Voice return. Alternative protection timings, including predictive pre-blocking armed from a learned burst cadence, were explored during development but showed no measurable benefit at the sparse-burst settings of this study and are not reported as a controlled ablation; timing remains an open axis under concentrated load.

Deployment constraints. Security against malicious or faulty Voice triggering (authentication, rate limiting, plausibility checks), finer-grained classes than a binary Voice/Best Effort split, and per-vehicle Best Effort fairness around the crash source are prerequisites before comparable MAC aggression could be defended outside a synthetic two-class study. Hardware-in-the-loop or testbed validation would further bound how much of the simulated protection-versus-cost frontier transfers to real 802.11p stacks.

In summary, crash-aware prioritization in decentralized vehicular Wi-Fi can measurably improve Voice delay and Voice-frame integrity where near-crash congestion is concentrated and bursts are dense enough to keep protection armed, but only as a carefully bounded, event-triggered escalation whose cost to ordinary traffic must be acknowledged and constrained.

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